GIANT LEAPS: GAP RESOLUTION IN SAFETY, NUTRITIONAL, ALLERGENICITY AND ENVIRONMENTAL ASSESSMENTS TO PROMOTE ALTERNATIVE PROTEIN UTILIZATION AND THE DIETARY SHIFT

Table 0.1. List of participants

Participant No.	Short name	Participant organisation name	Country
1 (Coordinator)	WR	STICHTING WAGENINGEN RESEARCH	NL
2	IRTA	INSTITUT DE RECERCA I TECNOLOGIA AGROALIMENTARIES	ES
3	MATIS	MATIS OHF	IS
4	UCC	UNIVERSITY COLLEGE CORK - NATIONAL UNIVERSITY OF IRELAND, CORK	IE
5	VTT	TEKNOLOGIAN TUTKIMUSKESKUS VTT OY	FI
6	UMan	THE UNIVERSITY OF MANCHESTER	UK
7	UNa	UNIVERSITA DEGLI STUDI DI NAPOLI FEDERICO II	IT
8	LUKE	LUONNONVARAKESKUS	FI
9	WU	WAGENINGEN UNIVERSITY	NL
10	DIL	DIL DEUTSCHES INSTITUT FUR LEBENSMITTELTECHNIK EV	DE
11	TG	TEAGASC - AGRICULTURE AND FOOD DEVELOPMENT AUTHORITY	IE
12	INRAE	INSTITUT NATIONAL DE RECHERCHE POUR L'AGRICULTURE, L'ALIMENTATION ET L'ENVIRONNEMENT	FR
13 (Affiliated)	APT	INSTITUT NATIONAL DES SCIENCES ET INDUSTRIES DU VIVANT ET DE L'ENVIRONNEMENT - AGROPARISTECH	FR
14	MUS	MEDICAL UNIVERSITY SOFIA	BG
15	AZTI	FUNDACION AZTI - AZTI FUNDAZIOA	ES
16	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES
17	EFF	STICHTING EFFOST	NL
18	B2F	FOODCOMPANIONS HOLDING BV	NL
19	VM	VIVA MARIS GMBH	DE
20	SF	SOLAR FOODS OY	FI
21	GCL	ASSOCIACAO OCEANO VERDE LABORATORIO COLABORATIVO PARA O DESENVOLVIMENTO DE TECNOLOGIAS E PRODUTOS VERDES DO OCEANO	PT
22	EM	EUROPA MEDIA SZOLGALTATO NON PROFITKOZHASZNU KFT	HU
23	FSN	FOOD SAFETY & NUTRITION CONSULTANCY	NL
24	Mosa	MOSA MEAT B.V.	NL
25	NFB	NAPIFERYN BIOTECH SP ZOO	PL
26	UL	UNILEVER U.K. CENTRAL RESOURCES LIMITED	UK
27	CAP	CAPACITY DEVELOPMENT IN NUTRITION RESEARCH-CAPNUTRA	RS
28	FC	FC CV	NL
29	DAAB	DEUTSCHER ALLERGIE- UND ASTHMABUND E.V.	DE
30	ROQ	ROQUETTE FRERES SA	FR
31	BD	RUKOV JAKOB LEWIN	DK
32	DNR	NUTRICIA RESEARCH BV	NL
33 (Associated)	ETHZ	EIDGENOESSISCHE TECHNISCHE HOCHSCHULE ZUERICH	CH
34 (Associated)	AGT	AGT FOOD AND INGREDIENTS	CA

1 EXCELLENCE

Summary

Accelerating the transition from animal-based to alternative dietary proteins – the dietary shift – is key to reducing the footprint of our food system in terms of greenhouse gas emissions (GHG), energy, water and land use, and other relevant environmental impacts, and for improving the health and well-being of people, animals and the planet. GIANT LEAPS delivers the strategic **innovations, methodologies**, and open-access **datasets** to speed up this dietary shift, in line with the Farm-to-Fork strategy and contributing to the Green Deal target of reaching climate neutrality by 2050. Achieving the dietary shift in practice is inherently complex due to the diverse set of actors involved and further hindered by major knowledge gaps ~~s~~cattered across the various alternative protein sources and the domains of *health* (safety, allergenicity and digestibility), *environment* (GHGs and other environmental and climate impacts, biodiversity, circularity), and/or *barriers to adoption* (technological, sensory, and consumer acceptance). The GIANT LEAPS consortium consists of the key actors and spans all expertise to address relevant knowledge gaps and proactively **engages** to arrive at **optimized future diets based on alternative proteins that are broadly accepted across stakeholder groups**. In order to deliver required insights for short-, mid- and long-term decision making and impact, GIANT LEAPS protein sources have been selected for either targeted or full assessment based on their current level of specification. The innovations and improved methodologies combined with accessible and comprehensive information generated for a wide collection of alternative proteins will enable:

- Policymakers to prioritise changes in the food system towards the dietary shift based on desired impact,
- Value chain actors to make strategic scientific, business and investment choices,
- The general public to make more sustainable and healthy dietary choices.

1.1 Objectives and ambition

Drivers for the transition

A transition from animal-based to **alternative protein diets is key to reducing environmental impacts and improving human health**. It has been estimated that food systems are responsible for 11.3% of total EU GHG emissions[1] and replacing the use of animal-based proteins will significantly reduce the dietary CO_2-equivalent impact of the European diet. The major impacts of the current food system on biodiversity, land and water use, and animal welfare, could be mitigated by a **shift from traditional animal-based towards more sustainable protein sources**. Alternative proteins have desirable nutritional characteristics and can contribute to consumer health as part of a balanced diet and reducing the incidence of chronic health problems associated with high consumption of traditional animal protein foods. However, alternative proteins can also negatively impact health, when contaminated with toxicological compounds or causing allergic reactions (food allergens are proteins and, hence, it needs to be ensured that new protein sources will not become new allergens), and therefore a safety by design approach is required.

Transdisciplinary approach

The human social and societal implications of the dietary shift are complex[2] and the necessary dietary transition will not occur until stakeholders from academia, industry, civil society and policy align. The GIANT LEAPS consortium is ideally placed to address this challenge, consisting of leading knowledge institutes, a combination of innovative SMEs and large food industries, not-for-profit organisations and partners with extensive experience in the public and regulatory space. The project takes a **co-creation, transdisciplinary and inclusive approach across the full range of supply chain actors** and with special consideration for **social well-being, including the health and safety of** consumers and vulnerable groups (e.g. people with food allergies, senior citizens or young children). These social and societal aspects must be balanced with economic considerations to achieve adoption of alternative protein diets in practice. Policymakers of European member states have expressed their food system sustainability ambitions, yet only a few have translated this into concrete plans, mainly due to a lack of scientific evidence about which alternative protein sources, technologies and policy instruments have the greatest potential impact. **GIANT LEAPS aims to engage with and provide information and solutions to all relevant stakeholders to enable the dietary shift**.

Knowledge gaps and results

GIANT LEAPS strategically **addresses major knowledge gaps relating to the protein transition in the 5 areas of: technological, health, safety, environment and consumer barriers to adoption** (see Fig. 1.1). It does so with **three concrete types of results**:

1. ***Innovations****: Identify and strategically address technological and sensory limitations and nutritional quality*

Substantial impact on environmental, climate and public health-related outcomes can only be achieved by large-scale consumption of healthy and sustainable protein sources, incorporated in food products that are accepted and regularly used by consumers: **the impact of the protein is in the eating**. Thus, impact will only be achieved if consumers develop preferences for these products, perceive claims about their nutritional and health properties as credible, and trust the underlying production chain. GIANT LEAPS strategically addresses technological, sensory, nutritional and

[1] European Commission, Reflection Paper, Towards a Sustainable Europe by 2030, 2019
[2] Santo et al. (2020), https://doi.org/10.3389/fsufs.2020.00134

other limitations that -to date -prevent (future) alternative protein sources from being used in foods and diets at scale, and clearly aligns product development strategies with consumer preferences

2. ***Methodologies**: Improve and innovate current methods*

Despite many existing methodologies and tools, critical methodological knowledge gaps exist within specific domains that limit relevant data collection, e.g. to successfully assess biodiversity impact in Life Cycle Assessments (LCA), predict allergenicity of novel proteins, assess protein digestibility and bioavailability without animal experiments, and model full impacts of diets on health and environment.

3. ***Datasets**: Gather data to create open access, comprehensive datasets*

A wide array of alternative protein sources is already available and even more will become available in the future. However, data related to health and environmental impacts, and critical applicability parameters (technological, nutritional and sensory functionalities) of these protein sources are scattered and incomplete. Therefore, GIANT LEAPS will develop a **future-proof, open cloud platform with protein screening AI algorithms** to collect and update comprehensive datasets from project-generated and existing data, covering a broad set of alternative proteins.

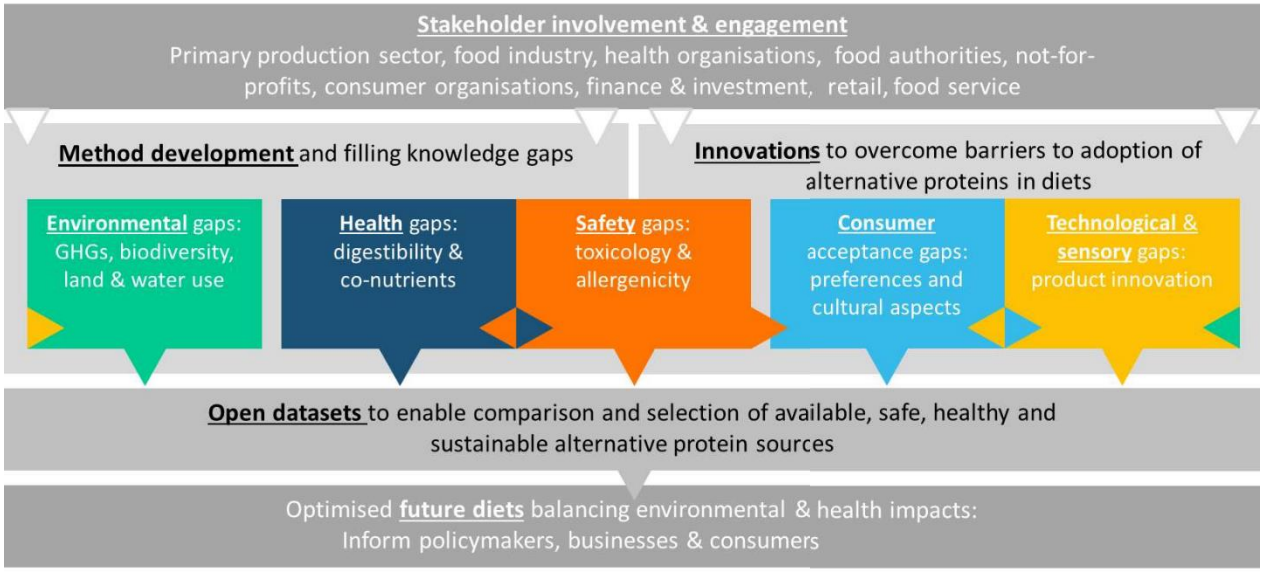

Fig. 1.1: GIANT LEAPS: stakeholders, knowledge gaps and results

1.1.1 Objectives

The GIANT LEAPS objective is to deliver the methodologies, datasets and innovations needed to **accelerate the adoption of healthy and sustainable alternative proteins and actualise a dietary shift across Europe**. The methodologies and datasets will be open (access) and comprehensive, providing stakeholders with a scientific basis to make future-proof and maximum-impact choices, from consumer-accepted policies to new technologies. To reach the overall objective, specific objectives (SO) have been defined, presented in Table 1.1, categorised as being primarily related to methodologies (M), datasets (D), innovations (I), and engagement (E).

Table 1.1. GIANT LEAPS Specific objectives

SO 1-E: Engage and co-create with policymakers, regulators, producers, businesses and networks, NGOs, consumers and consumer organisations to adapt or validate the approaches and specific choices (e.g. selection of protein sources) and perform consumer research providing insights to define societally acceptable future diets for four typical European areas taking into account regional food cultures, consumer habits and gender considerations.
SO 2-I: Deliver at least 6 food prototypes and technological innovations that optimally utilise proteins across at least 4 categories (plant, microalgae, insect, single-cell protein) with analytical results for health, safety and sustainability-related parameters.
SO 3-M: Deliver a safety by design toolbox, including novel cell-based tools for toxicological, allergic cross-reactivity and de novo sensitisation analysis, deliver 5 risk assessment scenarios for different classes of future alternative proteins, and develop consumer risk communication strategies and an online observatory for adverse response monitoring.
SO 4-M: Deliver an optimised and improved *in vitro* digestion method and standardised gut epithelial model that are validated by a human intervention study, assessment of a minimum of 9 alternative protein sources and formulated foods including bioavailability of amino acids, presence of co-nutrients and anti-nutritional factors, and *in vitro* methods to assess down-stream health effects.
SO 5-D: Develop an integrated sustainability framework including product environmental (PEF) category rules, construct and fill the Life Cycle Inventory database with sustainability data, and complete an assessment of 9 selected alternative protein value chains, compared to traditional protein chains.

SO 6-D: Deliver an operational cloud data platform, based on well-defined ontology with standardisation, interoperability and data requirements for maximal integration with existing EC-funded databases, available for data integration, analysis and exchange within the project and remaining available after the project ends (with Joint Research Centre (JRC) alignment and involvement).
SO 7-M: Optimized future diet: modelling framework fully defined, and the impact of the dietary shift – comparing the exchange of traditional animal protein with alternative protein foods – calculated and optimised for human health and planetary outcomes, for four regional diets in the EU.
SO 8-E: Development and dissemination of science-based policy briefs on alternative proteins as part of optimised future diets to support policymakers to successfully develop integrated and sectoral food-related policies, execution of an appropriate IPR strategy and exploitation plan for project innovations, and dissemination of main findings to all relevant audiences including the scientific community, farmers, producers and the general public.

1.1.2 Progress beyond the state of the art and ambition

This project is unique in that it is **comprehensive, transdisciplinary, and deeply interlinked**. The outputs of the project will advance science with new methodologies and innovative solutions, improve decision-making and policy through comprehensive shared datasets, and accelerate the transition through cross-cutting engagement. Earlier efforts have been focused on single protein categories, scientific approaches, or impact domains. **GIANT LEAPS will go beyond the state of art by applying a holistic approach, and maturing the development across protein sources and categories, encompassing safety, health and environmental assessment domains**.

*1. **Innovations**: Overcome bottlenecks with targeted innovations*

To achieve impact, dietary alternatives must be adopted in practice by consumers, i.e. alternative proteins foods should be produced and consumed at large across Europe. GIANT LEAPS will determine protein source/product combinations with high potential for consumer acceptance. Consumer research and impact estimation will be performed in four European regions (North, East, South, West), ensuring that the targeted future diets reflect regional habits and preferences, satisfy dietary guidelines, and offer culturally appropriate alternatives for consumers across the EU.

In developing the project content, partners and Stakeholder Board members have extensively discussed and evaluated protein sources, leading to a **preselection of protein sources for which knowledge developments are required on the short-, mid-, and long-term** (see section 1.2, Table 1.2 for more detail).[3] Animal and plant sources that are already extensively studied will be included only for purposes of creating comprehensive comparative datasets and for estimating future diets'impacts. Nine high-potential proteins with high potential to substitute traditional proteins **on the medium term: faba bean, lentil, oat, quinoa, rapeseed, microalgae, single cell bacteria, crickets and cultured beef,** will be characterized comprehensively and act as the focal point of processing and product innovation. Food prototypes and technological innovations for these proteins will be developed from TRL 3-4 to TRL 5-6, while safety assessment focuses mainly on **new, exploratory alternative proteins** (e.g. hemp, leaf proteins).

The **ambition of the GIANT LEAPS project** is to deliver a number of food prototypes and technologies that can be leveraged widely and have large potential for scale-up and industrial uptake to increase the availability of alternative protein-containing foods in the mid-term, contributing to the continued acceleration of the dietary shift.

*2. **Methodologies**: Advance scientific data collection with new and improved methodologies*

New methodologies will be developed to advance scientific understanding of environmental and health impacts (including safety risk assessment) of alternative proteins and foods and diets containing those, including:

a. Environmental assessment
- Set up comprehensive GIANT LEAPS sustainability framework, integrating existing and new methods.
- Develop innovative methods to assess **biodiversity, circularity and climate** potentials.
- Create new product environmental footprint (PEF) category rules for alternative proteins.
- Co-assess impacts on environment, natural resources, biodiversity and climate, acknowledging trade-offs.
- Combine existing environmental, social and economic data with new data to perform a holistic sustainability assessment, using predominantly Life Cycle Assessment (LCA)-based methods.

b. Health and safety assessment
- Set up a safety by design approach for **current and future** alternative protein sources, considering nutritional quality and bioavailability, and possible negative impacts (allergies, toxicological or other adverse effects).
- Develop and validate novel *in silico* and cell-based models to fill data gaps **without the need for animal studies (**e.g. replacing 90-day rodent studies), applied to selected protein sources, and assess the impact of ingredient and food processing.
- Demonstrate the safety by design approach with case studies for 5 different types of current and future alternative proteins with varying risk profiles.

[3] Selection of protein sources may be adapted based on stakeholder input (multi-actor approach) or state of the art progress before project start.

c. Assessment of dietary impacts on environment and health

- Define indicators for environment and health (nutrition, allergenicity, toxicity) and collect relevant data, and extend the SHARP (Sustainable, Healthy, Affordable, Reliable and Preferred) diet-model with these indicators (including at least land and water use, nitrogen and phosphorus)
- Assess impacts of alternative protein-based (future) diets on health and environment using the extended SHARP model taking into account cultural and regional dietary habits within four main regions in Europe, as compared to traditional animal protein-based diets, allowing realistic health and environmental synergies and trade-offs.
- Apply a multi-criteria decision analyses (MCDA)-like approach to allow rapid *a priori* analyses of the preferences and weights of various groups of stakeholders on the different types of health and environmental impacts of the diet, for use in the policymaking setting.
- Use optimisation techniques for environmental impacts of alternative protein-based diets while observing safety and other constraints.

The **ambition of GIANT LEAPS project** is to **develop improved, fast and animal-friendly methodologies** for the health, safety and environmental assessment of alternative proteins to arrive at future diets that are accepted by consumers and optimized for health and environmental impacts within the scope of the project, and to **disseminate these methodologies** to the scientific community so they can be applied widely to alternative protein sources now and in the future to keep accelerating the protein transition and the dietary shift.

3. Datasets: Accelerate decision-making with comprehensive datasets

No single holistic database on protein sources (traditional and alternatives) for food currently exists. Such a database would ideally combine multiple aspects of nutrition, health, safety and sustainability along the complete life cycle chains. The project aims to develop such a **holistic open-access data platform**. Existing ontologies will be integrated to combine available key databases on protein properties. This project links closely with other EC-funded infrastructures and initiatives, including the European Open Science Cloud, EFSA FoodEx database, SmartCLIDE and FNS-Cloud projects, for the purpose of developing a single framework. It will be aligned with the JRC Data Catalogue for long-term implementation. The data platform will be populated with existing and project-generated data, and can be further extended in the future. The integrated data platform will help identify existing data gaps, enable cooperation between project partners and direct future interdisciplinary work beyond the project.

The **ambition of the GIANT LEAPS project** is to **collect comprehensive data on 28 alternative proteins** (including traditional animal reference proteins, Table 1.2) to allow stakeholders to compare and benchmark proteins of interest and identify data gaps and targets per protein source. The open cloud data platform will remain an accessible and expandable resource after the project's end. This will allow scientists, industry and other end-users to define health and sustainability targets for new proteins and technologies to improve the state of the art, advancing the dietary shift.

*4. **Engage** to ensure additionality of work*

The four projects funded under the topic *LC-SFS-17-2019: Alternative proteins for food and feed* collectively represent state-of-the-art, each in their specific alternative protein domain. All coordinators of these projects are part of the GIANT LEAPS consortium and have been assigned roles as protein category Theme Leader. They coordinate the exchange of information and materials between the GIANT LEAPS activities and these projects. GIANT LEAPS will also connect to other relevant projects including *HORIZON-CL6-2021-FARM2FORK-01-02: Developing sustainable and competitive land-based protein crop systems and value chains* to ensure that GIANT LEAPS does not duplicate efforts but builds on and synergizes with emerging results.

The ambition of the GIANT LEAPS project is to engage with a wide range of stakeholders to ensure the GIANT LEAPS solutions will be taken up and exploited by relevant end-users, empowering them to progress the dietary shift. The overall **ambition of the GIANT LEAPS consortium is to provide coherent methodologies, datasets and innovations to achieve that 50% of total European protein dietary intake is derived from alternative protein sources by 2030**, as a steppingstone to reach levels of 60-70% by 2050. The ambition of **'50% by 2030'** will be made possible by advancing science, and informing and providing solutions to policymakers, businesses and the public to achieve positive health and environmental impacts.

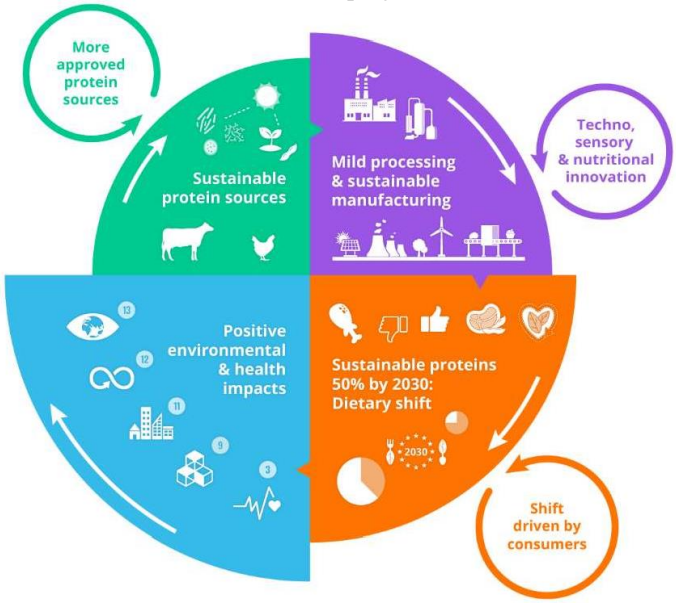

Fig. 1.2: GIANT LEAPS setup in relation to ambition and main impacts

1.2 Methodology

Engagement with stakeholders and end-users: a multi-actor approach

A **multi-actor and co-creation approach** (Fig. 1.3) is foundational to the GIANT LEAPS project and incorporated throughout project activities and subject domains. In addition to close collaboration with relevant running projects, an open **Stakeholder Board (SB)** has been convened and has already contributed to defining the main research questions of this proposal.[4] Engagement and co-creation will be facilitated in Work Package (WP) 1 (led by ETHZ), with involvement of all project partners to shape the research process of the project. For the interactive multi-actor an co-creation workshops (led by B2F), representatives from the following stakeholder groups will be invited: consumer associations, primary producers, food industry, retail/hospitality industry, public health authorities, risk assessors, and policymakers. The stakeholders will provide input on the project approach to ensure it delivers insights and solutions that are most relevant to stakeholders and to maximise impact collectively. The **SB will be extended before, during and after the project, resulting in a sustainable stakeholder platform** (see also section 2.2, Table 2.5).

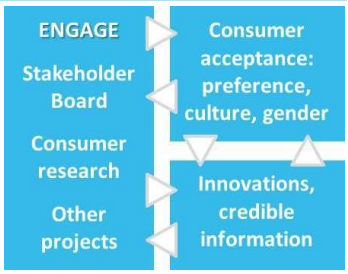

Fig. 1.3: Engagement overview

Past studies focused mostly on alternative proteins from single sources (e.g., insects, plant proteins (soy, wheat, corn, pea)) and lack the holistic approach. The proposed research will focus on a **broad set of 9 alternative proteins** which are not yet commonly applied in foods (lentils, faba beans, oat, quinoa, rapeseed, microalgae, single cell bacteria, insects, cultured meat) -although most of them are already on the market - to fill still-existing knowledge gaps (see also Table 1.2: **short list**), as well as on a selected set of novel alternative proteins for specific safety-related aspects in an exploratory way. This allows **identifying the highest potential alternatives for specific cultures and target groups in the different regions of Europe**.

Alternative proteins must be consumed as **replacement of traditional animal proteins**, not on top of those. This is of particular importance in Europe, where most consumers overconsume proteins mainly of animal origin. Culture is probably the most important factor driving people's food choices and behaviour.[5] Hence, it is important to better understand the **preferences, and drivers and barriers for consumer acceptance of alternative proteins in different European countries** through quantitative consumer research. For example, naturalness is an important factor influencing consumer perception,[6] and a perceived lack of naturalness may be a challenge for acceptance of alternative proteins in some countries and cultures. Consumer preferences, regional differences and food habits will be elucidated in WP1 in four regions across Europe, supporting the selection of producing food prototypes in WP2 and the estimation of impacts of accepted future diets in WP7. Considering feedback and input from consumers allows for balancing between sensory aspects, price, healthiness, naturalness, animal welfare and environmental friendliness.

Policies and regulations can dramatically impact the introduction of alternative proteins, and to a lesser extent on dietary patterns. This project examines and compares the **public acceptance of 'soft' and 'hard' policy measures,** from nudging to taxes. Using the generated data and methodologies as a science-based starting point for dialogue, the team engages with policymakers to develop and share best practices across countries.

Assessing alternative protein sources and benchmarking against traditional animal proteins

Development stage or TRL status of the various alternative protein sources with potential for food applications (plant-based, microbe-based, ocean-based, fungus-based, insect-based, and cultured meat) varies widely and, hence, **existing knowledge gaps vary significantly between different sources**. GIANT LEAPS brings together the main data generation and innovation efforts on protein sources (Fig. 1.4) with the highest potential for being newly or increasingly implemented in the European food chain. Table 1.2 (**short list**) presents these **highest-potential protein sources**. Most of them are already on the market but still only marginally used in our daily foods, whereas others are close to or have submitted Novel Food applications. These alternative protein sources will be addressed throughout all project activities, covering: ingredient processing and food production (WP2), safety and allergenicity (WP3), health and digestibility (WP4), sustainability (including environmental impact), biodiversity and

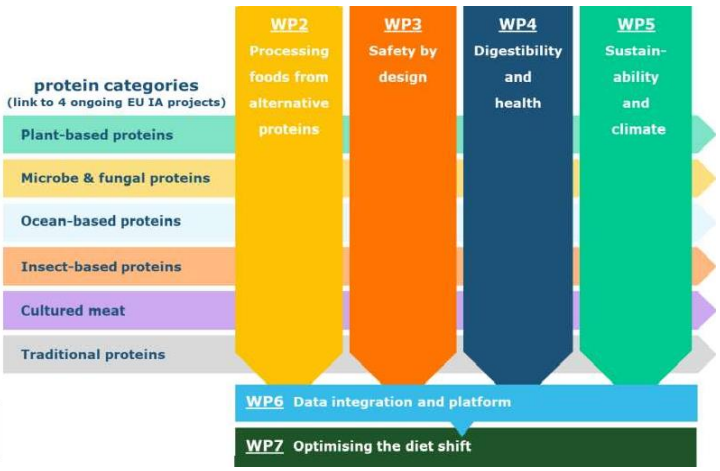

Fig. 1.4: Protein source analysis across GIANT LEAPS WPs

[4] A webinar with feedback session was organized on 8 September 2021 with representatives from 21 non-partner organizations.

[5] Rozin, P. (1996). Sociocultural influences on human food selection. In E. D. Capaldi (Ed.), Why we eat what we eat: The psychology of eating (pp. 233 263).
[6] Liñán et al. (2019), DOI: 10.12691/jfnr-7-9-10

climate (WP5), data integration and interoperability (WP6), and optimising the dietary shift (WP7). Existing data on these aspects will be collected and any missing data gaps be filled, to estimate the environmental and health impacts of future diets based on these alternative proteins (in WP7). Other alternative protein sources are in a **much earlier development stage** (**exploratory sources**, Table 1.2); WP3-4 will develop methodologies and case studies/series that will accelerate risk assessments, including data collection, and bringing those proteins to the market, creating long-term impact. For plant-based and other sources with few or no knowledge gaps (the **long list**, including traditional animal proteins as comparators, Table 1.2), existing data will be collected and analysed (in WP5-7). These sources will serve as a benchmark for project-generated data and will be incorporated in dietary modelling.

Table 1.2. List of proteins[7] categories, and strategies to achieve diet impact

Protein source selection[1]	Knowledge gaps and regulatory status	Strategy to achieve diet impact	WP	Impact timeline
Exploratory sources, e.g. hemp (P), leaf proteins (P), algal fractions (O)	Many knowledge gaps Regulatory approval missing	Improved RA methodologies to accelerate safety data collection of new alternative protein sources	3, 4	Long-term
Short list: faba bean (P), lentil (P), oat (P), quinoa (P), rapeseed (P), microalgae (Chlorella sp; O), single cell bacteria ('Solein'; M), crickets (I), cultured beef (C)	Specific / technological / sensory knowledge gaps Regulatory approval in place or pending	Technological innovations to increase availability of alternative protein-containing foods; assess environmental & health impacts of innovations; use data for comparisons and modelling optimised future diets	2, 3, 4, 5, 6, 7	Mid-term
Long list: soy (P), wheat (P), pea (P), chickpea (P), potato (P), brewer's spent grain (P), Spirulina sp. (O), fish side streams (O), krill (O); beef (T), pork (T), chicken (T), egg (T), milk (T), other dairy (T)	Few or no knowledge gaps Readily implementable in foods	Collect existing data for comparisons and for modelling approaches to define how alternative proteins can be optimally included in future diets for positive health and environmental impacts	5, 6, 7	Short-term

[1] *(P) plant proteins; (M) microbe & fungal proteins; (O) ocean-based proteins; (I) insect proteins; (C) cultured meat; (T) traditional animal proteins.*

Increasing the availability of foods containing alternative proteins

To increase alternative protein consumption in European diets, these must become more **available, accessible, affordable, and acceptable** to consumers. Processing technologies will be investigated to increase the availability of a diverse group of foods made of alternative protein ingredients (plant, microalgae, insect, microbe) by developing sustainable, healthy, and preferred alternative protein-based ingredients and food prototypes (Fig. 1.5).

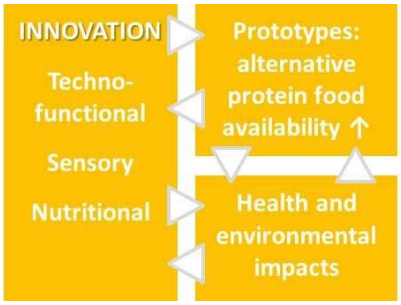

Fig. 1.5: Innovation overview

Effects of ingredient, food processing, and formulation will be investigated strategically (WP2, led by VTT) in relation to the key functional attributes, i.e. **nutritional, technological and sensory properties**, of the alternative protein sources in foods with complex structural hierarchies. The focus will be on **less refined, minimally processed ingredients** rather than highly refined ones to make use of multi-functionalities of distinct components (i.e. protein, dietary fibre, carbohydrates, minerals). Different protein sources will be blended (hybrids) to improve and complement the functional and nutritional properties of the end-products (IRTA, MATIS, UCC, VTT, VM, UL, FC). Applications in meat (MATIS, VTT, UL, VM, NFB, BD), dairy (IRTA, UCC, NFB, FC), egg alternatives (UCC) and culinary products (IRTA, MATIS) will be prioritised to maximise the potential impact of the innovations through substitution of these major dietary sources of traditional animal proteins. Ingredient and food processing, food formulation and matrix effects can have major effects on safety, health and environmental and sustainability aspects. Therefore, sample material and data generated in WP2 will be shared with WP3, 4 and 5 to: a) seek optimisation of these parameters through feedback and iterative steps, and b) include the impact of ingredient processing and food manufacturing on health and sustainability-related impacts (estimated in WP7) of future diets.

Ingredient-specific functionalisation strategies will be developed (IRTA, UCC, VTT, VM, NFB, UL, FC, BD, AGT) for improved technological functionality in food processing, as well as sensory characteristics and nutritional quality and safety aspects (allergenicity and microbial stability; interaction with WP3 and 4).

[7] Protein selection by be changed based on input from Stakeholder Board sessions, co-creation workshops or progress in the state of the art.

a) Available data for each raw material will be collected/updated to identify the most critical knowledge gaps.
b) **Functional properties of each protein ingredient will be characterised.** Particularly, protein composition, particle size distribution and application conditions (temperature, pH, ionic strength and solvents) will provide the essential background needed to understand protein-protein interactions and interactions with other polymers (i.e. dietary fibre, starch).
c) **Protein techno-functionality** aspects (such as solubility, emulsification, foaming, gelation, cohesion-adhesion, elasticity, viscosity, fat absorption, wettability, flavour binding and hydrophobic properties) will be investigated.
d) **Sensory properties** (flavour, off-flavour components, texture, taste) will be tested in suitable solutions. Proximate composition (water, protein, fat, ash) will be analysed, including dietary fibre (soluble vs insoluble) and mineral composition.
e) Various protein ingredients from diverse categories will be utilised in design of complex food model systems (from liquid to solid) which can be used as alternatives to animal protein-based foods.

Functionalisation approaches by thermo-, mechanical (VTT) (i.e. homogenisation, microfluidization, extrusion) and bio- (i.e. enzymes, fermentation) (IRTA, UCC, VTT) process technologies will be applied to achieve the **targeted nutritious and tasty alternative protein-based foods** *in situ*. A comprehensive evaluation of the functional properties of the modified proteins and the evolution of their structure conformation will be performed, to enable the functionalised protein sources to be applied in different food/beverage model systems (UCC). **Microstructure** (using CLSM, SEM) and **macrostructure investigation**, integrating rheology (oscillation and rotational tests) and tribology analysis (allowing to predict and understand the oral textural sensory properties of foods), and **design of new food structures** will enable the development of a powerful prediction tool for the product optimisation of the further range of food prototypes.

Healthy, sustainable and delicious prototype products for meat (wet & dry extruded, VTT), **dairy** (including drinks (UCC), yoghurt type gels (UCC), and cheese like spreads (IRTA)), **culinary products** (including patties, meat balls (MATIS, IRTA)) as well as **egg replacers for fine bakery** products (UCC) will be developed, based on the functionalisation strategies and food structure bioengineering innovations. **Consumer preferences/insight obtained in WP 1 will guide the selection** of ingredients and dedicated food applications. Food-compatible and label friendly technologies for improving the functional properties of protein ingredients will be addressed. Food prototypes will be characterised for rheological, textural properties, microstructure, flavour and mouthfeel, and selected samples will be sent to WP3 and 4 for further analyses.

Table 1.3 shows the short list of alternative protein sources with the highest potential for new applications in foods, that will be focused on regarding ingredient processing and food manufacturing. It concerns the **short list** (in Table 1.2), except for cultured meat because of (too) low volumes produced (for technology work in the project).

Table 1.3. Alternative protein sources addressed in ingredient processing and food manufacturing

Source, EU production volume and main countries	State of the art & enablers for food applications	Main barriers and knowledge gaps to address
Faba bean, 1.8 Million tons.[a] UK, FR, LT, DE, and IT.	Clear LCA assessment. Successfully applied for bakery products, extruded snacks and pasta. Some research done on nutritional functionality in foods.	Sensory issues (bitterness and astringency). Low performance in meat and dairy alternatives. FODMAPS and allergens, lack of knowledge on *in vivo* protein digestibility.
Lentil: 105.000 tons. ES, FR and EL.[a]	Clear LCA assessment. Successfully applied for beverages and pasta. Nutritional functionality studied mainly for beverage applications.	FODMAPS, lack of knowledge on *in vivo* protein digestibility and relationship between protein molecular structure and functionalities. Techno-functional and sensory challenges in meat and dairy alternatives.
Oat :13.6 Million tons. PL, FI, ES, DE, SE. [b]	Research focus mainly on fibre, less on protein. Fractionation designed mainly for β-glucan extraction. Applied for bakery products, extruded snacks and pasta, non-dairy milk.	Protein functionality is still inferior especially in meat alternatives (compared to legumes). Issues in dairy alternatives i.e. fermented products, linked to starch and/or dietary fibre.
Quinoa: 24511 tons in 2020.[c] UK, ES, FR, GE, IT	High nutritional quality (protein quality, gluten and allergen free, source of Fe, and zinc). Applied for gluten-free (fine)bakery, extruded snacks.	Sensory and techno-functional challenges for dairy and meat alternatives. Lack of knowledge in processing-protein functionality and *in vivo* protein digestibility.

Rapeseed: 23.2 Million tons DE, FR, PL, CZ, UK. [b]	Good nutritional quality (rich in fibre and protein balanced aa, higher Cys and Met than legumes). Sensory and/or techno-functional challenges. Tested in bakery products, meat alternatives.	Major sensory and techno-functional challenges in meat and dairy alternatives. Studies on nutritional functionality for food applications are limited.
Chlorella 82 tons/year (d.w.) PT, DE, NL, CZ, DK	High nutritional functionality (high protein content, source of fibre, all essential aa present, γ-linolenic acid, ω3 fatty acids and carotenoids). Used in a wide range of food products. Light Chlorella was recently established and is currently applied in a limited number of food products (e.g. sports bar, soups, crackers).	Techno-functional and sensory challenges in meat and dairy alternatives partially solved using light Chlorella (low chlorophyll content). Contrarily to wild type Chlorella vulgaris, the new light clones were recently established (2019-2020). Limited knowledge available regarding digestibility and safety.
Single cell protein (SCP) 10.000 tons in UK	New highly promising technology to produce bacterial SCP biomass from CO_2 with renewable energy. Amino acid and biomass composition comparable to other SCP sources. Bacterial SCP biomass is high in Fe and B-vitamins.	Techno-functional challenges especially for dairy and meat alternatives, Lack of knowledge in processing-protein functionality and *in vivo* protein digestibility.
Cricket (Acheta domesticus) 500 tons total EU insect products.[d] NL, BE, ES, LT, DK	Functional properties studied mainly in feed. Food related research is limited and focused on bakery products, minced meat, pasta and protein bars. Positive EFSA opinion on Novel Food status recently released– awaiting final EC approval. Complete protein with favourable amino acid profile and rich in essential micronutrients. High feed conversion ratio and wide substrate range.	Optimal application of cricket sensory qualities as well as product format (e.g. minced meat, protein bars, spreads and paste, cereal) with best market fit is yet to be identified. Sensory challenges through post-harvest drying and processing steps. Techno-functional challenges in integrating fresh or dried crickets in meat replacements.

Note: [a]Source: DG Agri, 2018; [b]Source: FAOstat, 2019; [c]Source: Mordor Intelligence 2020; [d]Source: IPIFF, 2019

Risk assessment (RA) strategies: Safety by design

In Europe, rigorous safety assessments considering microbiological and chemical food risks, impact on nutrition, and allergenicity, are applied and essential to ensure consumer health and trust in food. Such an assessment often comes at the end of product development and involves animal studies.

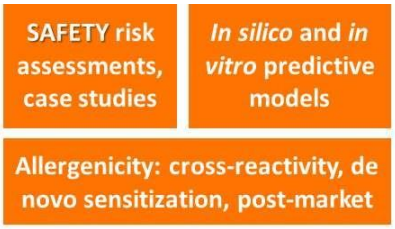

Fig. 1.6: Safety overview

a. **A *safety by design* approach is essential** to ensure that safety and health considerations are **embedded early and within the entire development of future alternative protein sources**. It needs to encompass both management of known safety hazards already subject to regulation (including upcoming regulation) and identification of any potential new hazard *(the unknown unknown')* – be it a new infectious hazard, toxin or new allergen. While the safety of the short list proteins (Table 1.2) has already (largely) been investigated, this is not yet the case for the exploratory protein sources (Table 1.2), and any other future alternative proteins. Additionally, changes in consumption – and hence exposure to potential hazards – that may increase due to the protein transition strategies also needs to be accounted for.

b. Animal models have been widely used to identify new hazards, but these are imperfect predictors of safety in a human context and in allergenicity RA. Classical toxicological studies, including genotoxicity assays and animal studies **have not been adapted to assessing dietary components such as proteins**. This is even more challenging in the context of allergenicity RA. The emergence of IgE-mediated food allergy as a condition of public health importance affects around 17 million Europeans.[8] This context, together with the societal drive to **reduce animal experimentation, requires further development of more effective and validated in silico and in vitro alternatives** with readouts that can be readily interpreted as to what might be considered safe and what may raise concerns.

c. **Effective post-market surveillance tools are lacking** to monitor unintended consequences of introducing a new food into the marketplace. Registries have been developed to collect data on severe, anaphylactic reactions reported by clinicians[9] and in iFAAM an on-line reporting tool was developed where patients could self-report their reactions.[10] However, no tools have successfully been developed to date to monitor allergic reactions to *alternative* proteins.

GIANT LEAPS will address the above-mentioned gaps (Fig. 1.6) by:

- **Setting up a safe-by-design framework including integrated effective set of *in silico* and *in vitro* tools** to support the safety assessment of future alternative protein ingredients, encompassing novel processing technologies,

[8] EFA report, 2019. Quality of Life for People with Food Allergies in Europe: A Menu for Improvement (LINK)
[9] Worm et al. (2014), doi: 10.3238/arztebl.2014.0367.

[10] Jusufagic et al. (2015), doi:10.3233/978-1-61499-564-7-320

enabling the rapid and safe introduction of alternative proteins into the marketplace. It will use data-driven approaches to methodology (including protocols) and reference data sets to support the assessment.

- **General risk assessment for a case-series**: The framework will be developed and worked out **or at least 5 different types of alternative proteins** (UMan) including both 2-3 proteins of a more developed nature (from short list: e.g. rapeseed and cricket flour) and 2-3 more experimental novel proteins (exploratory sources: e.g. from leaf and algal protein fractions). These will be benchmarked against foods of different nutritional quality (WP4) and allergenicity such as cow's milk, soybean and peanut for which knowledge has been built over many EU-funded projects, led by or with key involvement of GIANT LEAPS partners (Allergest, EuroPrevall, iFAAM, INFOGEST, IMPARAS, ThRALL). The selection will consider processing strategies ~~-~~both novel and conventional ~~t~~hat maybe used to mitigate or remove potential hazards such as chemical contaminants, microbes or allergens and include "hypoallergens" as a control, and thus feed into food processing mitigation strategies (WP2). Also, identified issues will be addressed through wider stakeholder engagement through WP1.
- **General safety assessment** (WR): Risks that lie in immunomodulatory effects of novel protein ingredients (including potential adjuvant effects of chemical compounds) will be assessed using human cell lines models of the gut epithelium (WP4) and with model immune cells. Immune cell differentiation and modulation of immune related pathways will be evaluated using (whole genome) gene expression analysis and cytokine profiling. A co-culture of immune cells and intestinal epithelial cells (IECs) derived from human induced pluripotent stem cells will emulate the *in vivo* intestinal environment and exposure conditions as accurately as possible. Immunomodulatory effects will again be evaluated as above and linked to assessments of potential genotoxicity in B-cells.
- **Allergenicity risk assessment for cross-reactivity to protect existing food allergic consumers** (INRAE): Existing *in silico* tools and databases will be aligned to facilitate phylogenetic and allergen sequence identity and similarity approaches together with motif searching, to incorporate HLA class II binding, using statistical and machine learning methods, to enable future initial screening of novel proteins. *In silico* predictions will be validated using IgE binding studies with serum panels from food allergic patients. Innovatively, this will be based on rational plasma selection and multiplexing or microfluidics, to reduce the volumes of precious biological samples.
- **Allergenicity risk assessment for *de novo* sensitisation** (WR)**:** *De novo* sensitisation, i.e. the capacity of an alternative protein to cause new food allergies, is less explored. GIANT LEAPS will develop models to assess the potential of alternative proteins to act as sensitisers, focusing on the gut, utilising gut epithelial models developed in WP4. Events in the gut *lumen* will be studied by using *in vitro* digesta applied to WP4 gut models, model proteins and alternative protein ingredients. Digestion peptide data sets will be used for modelling of HLA binding, again to facilitate initial screening. Events in the gut *epithelium* will be addressed mimicking either a healthy or an atopic, proinflammatory epithelium. This will be linked to *in vitro* methods to assess antigen-presenting cell (APC) and B-cell responses, to identify which parts of (novel) proteins can initiate immunogenic reactions. The capacity of APCs to activate naive PBMC-derived T-cells will be assessed and studies will be undertaken to investigate B-cell activation from human volunteers with and without atopic disease including food allergy. To validate *in vitro* results, *in vivo* sensitisation potential will be assessed in volunteers included in WP4 human intervention trial.
- **Consumer perspectives: risk communication and post-market surveillance in relation to food allergens** (UMan, DAAB): An outcome of allergenicity risk assessment is so-called precautionary labelling (implementing regulation (EU) 2021/120). Such statements may be mis-interpreted by allergic consumers and viewed as misleading and defensive. Based on patient-led approaches developed in the iFAAM project and the risks cross-reactive novel food ingredients pose to allergic consumers, effective ways of communicating those risks will be assessed. Post-market surveillance issues will be tackled by developing an electronic observatory based on the iFAAM reactions in the community project, to collect information on adverse reactions to alternative protein foods.

Digestion, bioavailability & health benefits

Digestibility generally describes protein quality: the better the digestibility, the higher the protein quality. The gold standard for analysis is the true ileal digestibility of each amino acid (DIAAS method), most commonly performed with pigs. Digestibility can be affected by many factors, e.g. protein structure, food matrix, food micro and macro structure, and physiological factors and life stages. Plant protein in its original food matrix (legumes, grains, nuts) is generally less digestible (<80%) than animal protein. Food processing has limited impact on animal protein but a profound effect on plant protein digestibility. Further, anti-nutritional factors (ANF), e.g. protease inhibitors, and co-nutrients like fats and fibre, can be affected by smart process designs.[11]

Fig. 1.7: Health overview

Digestibility, bioavailability, ANF and co-nutrients collectively affect health impacts of alternative proteins (Fig. 1.7).

Accurate assessment methods preferably avoiding animal models are needed to assess the digestive behaviour, bioavailability and overall digestibility of alternative protein sources and processed ingredients, and food products

[11] Prandi et al. (2021), doi:10.3389/fnut.2021.695793

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containing these. The INFOGEST model is an international consensus method, widely adopted by academia and industry, for the simulation of food digestion, protein digestion in particular, and is being evaluated as an alternative in vitro DIAAS ISO method. Proteins are digested to peptides in the stomach and the small intestine by the action of the pepsin and pancreatic enzymes (trypsin, chymotrypsin and carboxypeptidases). Finally, the brush border membrane (BBM) enzymes, located in vesicles on the epithelial cells of the small intestine, are the final step of the digestion, reducing peptides to free amino acids, di- and tripeptides that can be absorbed across the intestinal epithelium. So far, **no *in vitro* digestion method has included BBMs, and it the GIANT LEAPS ambition to improve the standard methods beyond the current state of the art by including these.**

Regarding *in vitro* models of gut epithelial barrier function, there are gaps in the knowledge of these polarised monolayers (e.g. Caco2-HT29MTX) when treated with food digesta, resulting in poor reproducibility and variable transport and permeability rates. Together with WP3, WP4 aims to harmonise protocols and define the characteristics of a healthy monolayer to **create a consensus protocol for gut barriers treated with food digesta**.

A human intervention study (DigestAA) will be performed, by one of the world-leading laboratories in the field, with two selected alternative proteins for accurate determination of the true ileal digestibility, which will also as a benchmark and validation of the new *in vitro* digestion method. Two alternative ^{15}N isotopes labelled protein-based foods, to distinguish food proteins from endogenous (secreted) proteins, will be assessed for digestibility in healthy volunteers. They will be equipped with a naso-intestinal tube the day before the postprandial test.[12] When the tube is located at the ileal site, samples will be collected continuously for 8h after the ingestion of the test meal, and breakdown products (amino acids and peptides) will be analysed, to determine the protein quality and metabolic bioavailability of dietary nitrogen. Also, an alternative, **novel and highly innovative method** will be used to indirectly quantify endogenous protein by prior administration of D_2O.[13] The determination of D-enrichment in individual amino acids allows to discriminate between endogenous (D-labelled) and dietary (unlabelled) amino acids. This method will be a **proof of concept to improve digestion studies**. The results of the human intervention study will feed into the validation of the new *in vitro* digestion method and WP3 safety assessment (characterisation of released peptides and potential epitopes).

Nutritional passport: Alternative proteins (**short list**) will be tested for digestive behaviour, bioavailability and health biomarkers to develop a nutritional passport, including the presence of co-nutrient information, allowing comparison with reference proteins (soy, egg or whey). Mass spectrometry mapping of the digestion products - especially on the basolateral side of the epithelial cell layer- will be applied to identify peptide sequences with beneficial health effects and confirmed with cell-based assays. *In silico* prediction of bioactivity will also be addressed. WP4 will collate and bridge the knowledge gap for WP2-selected proteins, and -in interaction- will investigate specific processing steps to improve digestibility and digestive behaviour.

Holistic sustainability assessment environment, society and economy

The EU food system cannot currently be considered sustainable with respect to **environmental, social and economic aspects**.[14] The way in which we produce and consume animal proteins is causing significant negative environmental impacts. Plant sources which currently dominate the global protein supply also cause issues: land clearings, intensive use of croplands, habitat fragmentation, threats to biodiversity, GHG emissions and fertiliser use. As a large proportion of these are currently converted into animal protein for human consumption, the environmental pressure derived from traditional animal protein production is further intensified. From the broadest perspective possible, environmental impacts of alternative proteins do not give a correct estimation of their sustainability. Human well-being is one of the key aspects of the long-term sustainable development of the planet, with fourteen of the seventeen UN's SDGs dealing with social impacts[15]. **Biodiversity loss** has been identified as one of the three most critical environmental impacts caused by human activities, and biodiversity protection and restoration is one of the 6 environmental objectives of the European Union (EC/2020/85).[16] Nevertheless, many mainstream LCA methodologies have largely ignored biodiversity loss.[17] Active development has been in progress to include biodiversity as an impact category in LCA but significant methodological challenges still remain. To fill the evidence base, pros and cons of producing alternative proteins on sustainability, biodiversity and climate (neutrality/adaptation) need to be assessed, in comparison to traditional animal proteins. GIANT LEAPS will address this (Fig. 1.8) as follows:

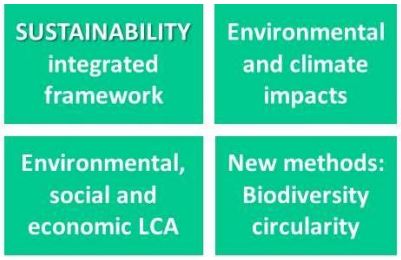

Fig. 1.8: Sustainability overview

- An **integrated GIANT LEAPS sustainability assessment framework** will be developed (by ETHZ) to conduct a sustainability assessment of the alternative proteins, considering all three pillars of sustainability: environmental,

[12] Oberli et al. (2015), doi:10.3945/jn.115.216838; Calvez et al. (2020), doi:10.1093/ajcn/nqaa274
[13] Tessier et al. (2020), doi: 10.1093/jn/nxz279
[14] European Commission (2020). Towards a sustainable food system.
[15] Source: CAP Context indicator C.26 on Agricultural entrepreneurial income (https://agridata.ec.europa.eu/Qlik_Downloads/Jobs-Growth-sources.htm)
[16] Rockström et al. (2009), https://doi.org/10.1038/461472a

[17] FAO (2020) Biodiversity and the livestock sector - Guidelines for quantitative assessment: Version 1. FAO, Rome, Italy

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social and economic impacts. The framework will be based on LCA and Social Life Cycle Assessment (S-LCA), and Environmental Life Cycle Costing (E-LCC). The sustainability assessment framework will **go beyond state of the art by implementing results from novel biodiversity impact analysis and analysis of circularity potentials of the alternative proteins** (by DIL), offering a broader sustainability profile and **by setting the ground for new PEF based categories for alternative proteins**. Costs and benefits (WU) of the selected alternative proteins for climate and biodiversity will be incorporated into the framework, using LCA and non-LCA methods. Methods for each sustainability dimension will be defined in the framework based on scientific methods following recognized European and international standards and frameworks. The framework will then be used to build the comprehensive sustainability models of the alternative protein systems.

- **Environmental sustainability: Impacts of selected alternative proteins on all five main drivers of biodiversity loss** will be assessed. LUKE will apply existing LCA methods to cover three of the main drivers (habitat change, climate change and pollution) while separate (non-LCA) indicators will be used to assess impacts on the remaining two drivers (invasive species and overexploitation), for which no operational LCA-methods yet exist. This innovative approach will be integrated into the sustainability assessment to fill knowledge gaps on the environmental footprint of alternative proteins.
- **Social sustainability**: GIANT LEAPS will assess social impacts (LUKE) through Social Life Cycle Assessment (SLCA), an approach that was formalised by the SLCA Guidelines by UNEP/SETAC, updated in 2020. The method considers the entire value chain of products and the multitude of stakeholders found along it, including the workers in the fields and in the factories, the consumers, the local communities impacted, other value chain actors and society as a whole. The results are often site-sensitive; thus, primary data will be collected through questionnaires and focus groups to reflect the **reality of production of alternative proteins in several areas in Europe and the potential switch from one protein type to another**. The aim is to better understand the positive and negative social impacts and to ensure that all project implementation activities have, as much as possible, a positive effect. The lessons learnt will be of great importance for future agricultural projects and for companies looking to produce sustainable alternative proteins.
- **Economic sustainability**: A Life Cycle Costing approach (WU) will be applied to assess the economic sustainability of the production chains of alternative protein sources. This assessment focuses on the internal costs of the production chain to evaluate its economic viability. Scenario analyses will be performed to account for impacts due to potential changes in the global protein market situation.
- **Climate adaptation/mitigation:** Results from the LCA analysis will be further used (by MATIS), and an additional protein vulnerability analysis be done to **provide detailed climate profiles of the alternative proteins**. This will indicate the alternative proteins potentials to serve as climate mitigation and/or adaptation methods, and also give stakeholders an insight into key climate vulnerabilities of the production of the proteins, which, if addressed, can largely enhance their resilience towards main expected climate risks.
- **Data collection**: The extensive amount of data from various sources collected and generated in WP5 throughout the project's lifetime will be accumulated (by DIL) in a Life Cycle Inventory (LCI) data repository, which will be freely available for research purposes, and linked to the cloud platform in WP6. This LCI repository will be used in WP5 sustainability impact assessment and thus serve as an important tool in filling existing knowledge gaps on the environmental, social and economic impacts of traditional and alternative proteins sources.

This comprehensive and comparative analysis will provide stakeholders and policymakers with new, detailed knowledge on the potential impacts of environmental, societal and/or economic changes on the production of the alternative proteins and assess how their production and consumption may affect our society, economy and environment, including biodiversity and climate, and make a **clear comparison with traditional proteins**.

The sustainability analysis of WP5 is directly linked with WP6 regarding collecting and delivering data on sustainability aspects according to the defined methodology of data collection and management. WP5 will use data from WP2, WP3 and WP4 in their environmental, social and economic analyses. The results of all analyses performed in WP5 will feed directly into WP7, where the sustainability of the future dietary shifts will be analysed.

Integration of data and impact of future diets

To provide policymakers and other decision-makers in the food system with all relevant information to plan and achieve the protein transition successfully, it is paramount that all the types of data streams related to the use and processing of protein sources in diets, health risks and benefits, and sustainability are combined and integrated into one open assessment framework (Fig. 1.9).

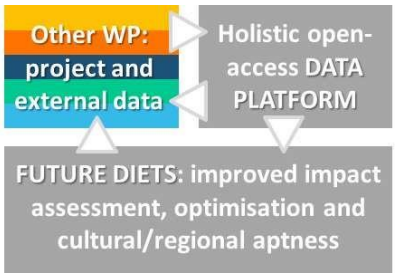

Fig. 1.9. Data platform and future diets overview

An open access data integration and interoperability platform will be developed in WP6 in order to bring all data streams together (including project-generated and external data), to provide an overview of available and missing data per protein source and to **create a future-proof data infrastructure** that can be used by scientists and policymakers. The open cloud data platform

addresses the gap of data availability on protein sources (including alternative) holistically combining multiple aspects of nutrition, safety, health and sustainability along the complete life cycle chains. It will serve as the **one-point of entry** for integrative modelling by relevant stakeholders, policymakers and scientists, e.g. performed in WP7 for modelling and optimization dietary impact.

Data integration protocols will be defined in interaction with WP2, WP3, WP4 and WP5 and public data on protein sources will be captured. The platform will support data discovery and reuse, methodology and **open frameworks** to facilitate curation and annotation with community standards (e.g. ISA) and established metadata that can be explored and annotated according to FAIR principles and relevant disciplines through interpretational module of developed ontology (WP6). Data semantics will be structured hierarchically and stored in an ontology, together with their relationships for formulating data search strategies and interoperability. Furthermore, a new **explainable Artificial Intelligence approach** (AZTI) will be applied to predict the characteristics and impacts of alternative proteins.

GIANT LEAPS' data platform will be built to integrate aggregated data into **open access cloud databases** for scientific purposes, in cooperation and alignment with external EC-funded projects, initiatives and infrastructures, including FNS-Cloud, SmartCLIDE projects and the European Open Science Cloud. Active involvement from JRC will be sought to ensure maximal interoperability. Overall, it aims to activate data for interdisciplinary work (e.g. for the application in WP7-8) and cooperation between project partners and beyond, extending it to external users.

The overall impact of the dietary shift towards alternative protein foods

The European food system has the potential to operate in a more coherent and efficient manner to work towards common health and planetary goals. Dietary patterns representative of all European regions need to be estimated, especially pertaining to protein. The **innovative inclusion of health metrics** (allergenicity, quality, safety) **and environment metrics** (nitrogen, greenhouse gas emission, land use, water use) will mean significantly broader applicability and exploitation of these models. GIANT LEAPS will use the available building blocks and extend the current models with health and environmental metrics.

a. **Indicators for health and environment will be established**, that are quantifiable and can be normalised in order to combine them in an overall output.
b. Relevant data on the indicators will be collected from the other WPs, available databases and published studies (WR, WU), as aggregated in the data integration and interoperability platform (WP6), and **incorporated in the GIANT LEAPS dietary shift modelling approach**.
c. The impacts of **typical current (traditional protein) diets** across Europe on the health and environmental metrics will be estimated using this framework, for updated **dietary intake data from 16 countries, averaged across 4 European regions** (N/S/E/W).
d. Based on the typical diets, also **future ~~a~~lternative protein-based diets ~~w~~ill be composed** for each of the fourEU regions, taking into account regional dietary guidelines, culture and eating habits, and **impacts of these futurediets on health and the environment will be assessed** as well. This extends the current SHARP and SUSFANS modelling approaches to include food safety, allergenicity, bioavailability, nitrogen and phosphorus, and water and land use impacts, as well as to include alternative protein-based foods.
e. Preferences of different stakeholder groups will be included, using a Multi-Criteria Decision Analysis (MCDA), an approach that has successfully been applied to compare potential mutually exclusive factors.[18,19] Each of the indicators can be weighed equally or weights can be attributed to the various indicators depending on their perceived importance. The outcome of such an MCDA allows **fast *ex ante* comparison of various future diet scenarios** on health, sustainability and other relevant factors.
f. **Alternative protein diets will be optimized**, using (multi-objective) optimization modelling to minimize negative environmental impact while not reducing (or increasing) consumer health

The GIANT LEAPS modelling framework will be used in interaction with input from stakeholders, with results providing input to other WPs (see Interaction with other WPs), in this way providing a feedback loop.

1.2.3 National or international research and innovation activities

GIANT LEAPS will build upon many past and ongoing national and international projects, presented in Table 1.4, and will focus on integrating the knowledge, experiences, tools and innovations gained within the separate domains of research. The specific partner involvement per project is described in part A of the proposal.

Table 1.4. National and international projects and activities categorised by topic area.

Topic and relevance to GIANT LEAPS	Projects with GIANT LEAPS partner involvement
Many partners are or have been involved in projects around **specific classes of alternative and sustainable protein sources**, including dedicated HORIZON 2020 and national calls	EU-SUSINCHAIN and InUrban (Insects); EU-ProFuture (Algae) EU-NextGenProteins (diverse selection); EU-SMART PROTEIN (terrestrial); EU-Plenitude (Fungi); EU-BBI-PROMINENT (cereal side streams); Sustainable Future Proteins (diverse selection); EIT

[18] Rosa et al. (2016), doi:10.1016/j.ecolind.2014.02.027
[19] Burkhard et al. (2014), doi:10.1016/j.ecolind.2014.02.022

focusing on sustainable proteins. A limited selection is shown here.	FOOD Taste2Meat (sunflower); SUSFOOD-OATPRO (oat); EU-AQUABIOPRO_FIT (fish-side stream); EU-FASTTRACK (oil plants/ rapeseed); EU-GREENPROTEIN (leaf protein)
The GIANT LEAPS consortium can build on the **allergenicity** expertise developed within Europe, notably due to several large EU projects that partners led or contributed to.	EU-iFAAM (prevent and manage); EU-EuroPrevall (information tools for authorities and industry); PREFER (toolbox for sensitisation); ThRAll (detection); COST-ImpARAS (allergy risk assessment strategies)
Partners have been involved in research and activities related to **risk assessment of food** and were/are engaged in **novel food procedures** and food legislation.	EU-NOFORISK; EU-MyToolbox (mycotoxins); EU-GRACE (maize); ALGAESOLUTIONS (algal products); Support to several EFSA dossier submissions (Insects, Duckweed, Beet Leaf, and several ongoing); Activities as panel member of EFSA
Digestibility & health, important quality characteristics of proteins and foods, were the topic of several projects the partners were or are involved in.	COST-INFOGEST (*in vitro* digestion methods); VistaMilk (dairy); TOMI and InfantDigest (infant formula); EAT4AGE (food for elderly); SOLARBIOME; PRODIGE (oilseed proteins); SustainableFutureProtein (alternative proteins)
Innovations in product development have been developed and ongoing projects, mainly on a national level. The GIANT LEAPS will connect expertise and accelerate product development by combining this knowledge.	PlantMeatMatters, PlantPromise and EXPRO (meat replacers); FUNGITIME (fungi biomass-based food products); EU-BBI-PROMINENT (cereal sides streams); EU_ALSEOS (rapeseed waste to food); EU-FIDOs (products from oilseed co-stream); Fifteen crickets (products from crickets)
Innovations in production processes to increase quality, functionalities, cost efficiency and sustainability were topics addressed in several large projects and many smaller bilateral projects not listed here.	EU-Enthalpy; EU-NewTechAqua; EU-MASTER (fermentation); EU-PROTEIN2FOOD; EU-APROPOS (biomechanical processing for rapeseed press-cake and fish side-streams); EU-GREENPROTEIN (protein extraction); PARP (protein extraction); BRIdgeALfa (technologies for rapeseed and soy)
Life cycle analysis, sustainability of proteins and agri-products were specifically addressed in these projects.	EU-ClimeFish; EU-CLIMAQUA; SEFARI (pathways to sustainable protein commodities); BIOEASTsUP (circular economies); LIFEinFORESTS and EU-BIOVALUE (biodiversity)
Consumer aspects, like trust, human diet and relation to health and sustainability were included in several projects listed above but addressed explicitly in the projects listed here. Many national-level projects are omitted.	EU-SUSFANS (metrics, models and foresights for sustainable food and nutrition security Europe); EU-EFCOVAL (Food consumption validation); EU-EURRECA (micronutrient recommendations); EFSA RPC (models for consumption data); SUP-DIE (consumption habits); EU-SMART PROTEIN (Trust)
GIANT LEAPS will continue and integrate **multi-actor engagement** as initiated in several projects, including those listed here.	EU-RADIANT; TRADEIT (regional competitiveness SMEs); INNO4AGRIFOOD (SME Agri-Food ecosystems); MY-GATEWAY (boosting start-ups)
GIANT LEAPS will use the **data sets and platforms and infrastructure** developed in past and current projects. Projects also include IT solutions and AI approaches to support data integration, data analysis and exploitation.	FNS-Cloud (data-knowledge and tools); CoEICT (AI for allergy prediction); ICTSES (immunotoxicology); ProteinCompass (database on protein functionality and nutrition); P-PRO (protein processing ontologies); TECNOMIFOOD (AI for precision nutrition); ASSIST and DIY4U (Digital platforms)

1.2.4 Integration of disciplines

The GIANT LEAPS scientific approach and assessment methodology to optimise sustainability- and health related outcomes of future alternative protein-containing diets integrates, by design and necessity, a wide range of scientific disciplines. **Diverse fields are brought together, including food science and technology, social sciences and humanities, life sciences, environmental sciences, and data sciences including modelling and artificial intelligence**. Considering **all aspects of the value chain**, a systems-oriented approach is necessary to realise the ambition to reshape the food production, processing and consumption and bring this to the next level. As is illustrated in Fig. 1.1 (section 1.1) and in the WP structure diagram (Fig. 3.1, section 3.1), the integration of disciplines is at the heart of GIANT LEAPS. In addition, it is the explicit purpose of WP6 (open data integration platform) to enable the integration of data stemming from the wide variety of scientific disciplines and associated methodologies.

Many of the GIANT LEAPS partners have extensive experience with integrating scientific disciplines during research activities. For example, LCA, socioeconomics and studies on climate impact and dietary shifts integrate various disciplines that require a broad knowledge and understanding of the agro-domain, land use, crops and production, processing, food consumption etc. During project meetings, the interaction and information exchange between partners, WPs and disciplines will be encouraged through **plenary sessions and joint co-creation workshops** with a wide range of stakeholders, including policymakers, primary producers, food producers and consumers.

1.2.5 Integration of social sciences and humanities

GIANT LEAPS strategically addresses alternative protein-related knowledge gaps with the end goal to achieve a dietary shift and sustainable food system. The success of the dietary shift and the environmental and health impacts of this transition critically depends on the **willingness of consumers** to consume alternative protein-containing foods as integral part of daily meals and on the **economic feasibility** of the change in the food system for all actors across the value chain. Therefore, these two critical aspects related to human behaviour are an integral part of GIANT LEAPS. Within WP1, quantitative consumer research is performed, tailored to 4 regions across Europe, to address regional and cultural differences, to ensure that the innovations that are developed in WP2 and the future diets that are defined in WP7 align with consumer perspectives to maximise the chance of success and long-term impact of the dietary shift. WP1 also defines policies to increase alternative protein food production and consumption that is acceptable to consumers. Within WP5, economic and social aspects are assessed as part of a holistic sustainability, climate and biodiversity assessment. The essence of the GIANT LEAPS proposal is the i**ntegration of these social sciences and humanities disciplines with technology, life science and health-oriented disciplines,** combining the related insights and information flows to define optimised future diets that can guide policymakers, businesses and consumers towardsa dietary shift, sustainable food system and positive environmental and health outcomes.

1.2.6 The gender dimension (i.e. sex and/or gender analysis)

GIANT LEAPS will actively address the gender dimension in its activities throughout the WPs by facilitating inclusion of sex and gender as critical analytical and explanatory variables in its research and involvement of the multi-actor stakeholder interactions. In all proposed studies, measures will be taken to ensure the different genders will equally participate. Gender analysis and gender dimensions are included in all relevant project activities to enable a shift towards healthy and sustainable diets, supporting climate change and transforming the food system in a fair and equitable way.

Statistics show that **females have a major influence on the daily meals** of European households,[20] and are also more **concerned about sustainability of food production and climate** issues than male.[21] Regarding alternative proteins, the influence of gender seems to vary, as insects are more accepted by males and plant-based proteins by females.[22] Therefore, gender is an important variable in the consumer surveys (WP1). Information about gender will be collected and used in the analyses. A non-binary answer option will always be included but, based on past studies, it is not expected that a sufficiently large number of people will use the option for it to be included as a category in the analyses. Therefore, the main comparison will be between males and females. The sample will consist of equal numbers of females and males for each consumer study, ensured by using quota samples. Within WP2, no prototypes or innovations are developed that favour either female or male end users, but input from consumer research in WP1 will ensure that gender dimensions are addressed in identifying the best opportunities to develop new food prototypes. In WP3 and WP4, sex-specific dietary and nutrient responses in relation to food safety and health are considered, with special consideration for vulnerable groups including pregnant and/or breastfeeding women. The human study in WP4 is performed on healthy volunteers and the inclusion criteria are defined broadly to represent the healthy adult population: men and women, aged 18 to 65 years, BMI 18-30. The aim is to include males and females equally. Within WP5, gender dimensions will be part of the social LCA assessment, and in WP7 future diets for the four regions in the EU will be defined, ensuring that these are well-balanced from a gender perspective. The GIANT LEAPS consortium will support the inclusion of gender perspectives in all its policy recommendations (WP8), in line with the Gender Equality Strategy 2020-2025, which aims to "redistribute power, influence and resources in a fair and gender-equal way, tackling inequality, promoting fairness, and creating opportunity".[23] Moreover, recognising the importance of battling gender stereotypes, communication and dissemination activities of WP8 will avoid the use of stereotypical images and messaging in relation to gender and where relevant, challenge any norms to bolster shifts in people's beliefs and values towards gender equality.

Through new knowledge and innovations, WP8 will identify pathways to support the creation of new jobs and preparing the workforce for jobs of the future in the food and bioeconomy sector with an emphasis on training/capacity-building activities to improve professional skills and competencies. Strategy and work on this will aim to **support equal participation and payments of men and women** across the relevant sectors of the economy.

1.2.7 Open science practices

GIANT LEAPS will produce tremendous amounts of data, publications and other scientific outputs. To ensure the highest possible quality, efficiency and responsiveness of the research, **Open Science is the standard way** of working within GIANT LEAPS, and integral to the exploitation and dissemination plan. The following principles apply:

[20] Eurostat (2019) Participation rate in cleaning and food management, by gender, % (2008 to 2015)
[21] OECD (2014) https://doi.org/10.1787/9789264214651-en
[22] Onwezen et al. (2021) https://doi.org/10.1016/j.appet.2020.105058
[23] European Commission, *Gender Equality Strategy 2020-2025*, p.2; European Commission, DG for Research and Innovation (2013), *Gendered Innovations: How Gender Analysis Contributes to Research*, pp.26, 28.

- All publications will be made openly available and shared as early as possible within the publication process. Popular articles will be shared, in addition to the print versions, on the GIANT LEAPS website and websites of the involved partners. Peer-reviewed scientific publications will be submitted to **Open Access journals**, with pre-prints published (e.g. at ResearchSquareF100). Protection of knowledge will be ensured by adopting licenses that enable free circulation of documents while safeguarding authors' (and the project's) IPR.
- **Open data sharing** principles: All data collected within the project will be shared with the community, wherever possible, applying **FAIR** (Findable, Accessible, Interoperable and Re-usable data) principles. Data will be shared with the public repositories, or as separate datasets, e.g., on data repositories like European Open Science Cloud (EOSC), Zenodo or FAIRSharing.
- GIANT LEAPS researchers adheres to the commonly agreed standards of **research integrity**, i.e. the proposed research is conducted so that others have confidence and trust in the methods and results. All research activities, such as experiments, modelling etc, are defined and executed in such a way they are reproducible and re-usable. Research hypotheses will be stored on beforehand on OSF (https://osf.io/prereg/) or AsPredicted (https://aspredicted.org/).
- Involved scientists are supported by the **WP9 Open Science and Data Manager, as well as Data Stewards** of own organizations) to apply proper Data Management and Open Science research routines and practices. **Mutual learning exercises and training** are implemented for exchange of experiences and learning from each other. Open science practices will be linked to innovation and business models, by involvement of the **WP9 Innovation and IPR (IIPR) manager**, requiring careful consideration of issues like IPR, licensing agreements, interoperability, and data reuse.
- European consumers and citizens are recognized by GIANT LEAPS as **valid contributors** to achieving the scientific aims of the project. **Citizens, consumers and civil society organisations and end-users will be involved** in the co-design and co-creation processes, particularly in the three multi-actor WP1 workshops, and consumers are involved in the WP1 surveys to guide the selection of ingredients for dedicated food applications in WP 2 and to obtain realistic and accepted future diets in WP7.

1.2.8 Research data management and management of other research outputs

Data Management is an integral part of the GIANT LEAPS research process. At the start of the project, a detailed **Data Management Plan** (DMP) will be developed, using the Horizon Europe template, by the **Open Science and Data Manager (OSDM)** in agreement with all partners. This DMP will be delivered (in WP9) at Month 6, and will be updated every 6 months by adding more detailed information, such as applying relevant ontologies and data format standards (for e.g. protein composition and sustainability and health indicators), and providing updates as the project progresses. GIANT LEAPS applies **FAIR (Findable, Accessible, Interoperable, Reusable) principles**, implying that data are as open as possible and as closed as necessary. The DMP will outline how these principles are applied in practice in our project. Since already existing data will be gathered and (re-) used in the project, particularly in WP5, 6, and 7, the DMP will also cover these re-used datasets.

FAIR principles: Newly generated data will initially be stored locally in the institutional repositories of the respective partner. Metadata of all original datafiles, compliant with the Horizon Europe guidelines, will be made publicly available. Then, newly generated data will be made openly accessible in trusted public repositories, such as EOSC, whenever possible, while keeping robust IPR management (data linked to IP will be available after only IP protection mechanisms have been put in place), and with the metadata standard of that repository added to the dataset. Source code, and AI and statical models will be made available in open source version control systems, like GIT, to make the analysis reproducible (e.g. R or Python Notebooks), let others review it and share it publicly.

***Other research outputs*:** The DMP, next to data, will also include a plan on how other research results that will be generated by or re-used in the project will be handled. These include, amongst others, processing technologies in WP2, gut models and bio-assays in WP3, nutritional passports for alternative proteins and *in vitro* digestion models as alternatives to animal models in WP4, Life Cycle Inventories and system models in WP5, AI algorithms in WP6, and the SHARP model in WP7.

***Resources*:** Each partner in the project has allocated costs (mainly personnel) for making data FAIR and for a data steward. The partners' **Data Stewards** monitor the data collection process of the respective partner, make sure FAIR principles are abided throughout the duration of the project, and provide assistance whenever needed. The project's responsibility for data management is with the project **Open Science and Data Manager**, but ultimately with the PC, and the aspect of proper data management will be covered as standard agenda point of the annual consortium meetings and the ExC meetings.

***Data security*:** Data storage and security will be managed by the respective institutions, such as their local drives and security measures. A part of the collected data is sensitive and will be processed in accordance with regulatory (GDPR) and institutional guidelines. This includes a data privacy assessment, data minimization, providing an informed consent form and anonymization/pseudonymization. Further, sensitive data will be stored in password protected folders and where possible encrypted. Data ownership and sharing of the data will be covered in the consortium agreement of this project. Datasets, where restrictions (e.g. privacy) apply, will be archived on the

network drive of the party who owns the data and be available for 10 years, in accordance with the VSNU code of conduct for research integrity.

***Regulatory and Ethical aspects*:** GIANT LEAPS will ensure full compliance with ethic regulations and GDPR (General Data Protection Regulation) for personal data gathered in the project, particularly the stakeholders in WP1, allergenicity work in WP3, and the human trial of WP4. In the WP1 consumer surveys, only fully anonymized and no personal data will be collected. More details are given in the ethics self-assessments of involved WP1, 3 and 4 partners. Ethical approvals will be requested for at the responsible committees prior to the planned research.

2 IMPACT

GIANT LEAPS offers a ground-breaking integrated approach to addressing pathway 4 – alternative proteins and dietary shift ~~-o~~f the Food 2030 pathways for action.[24] It contributes to the Green Deal priorities, Farm to Fork strategy and EU's climate ambition by reducing emissions, improving health and wellbeing, and transforming the food system towards sustainability and circularity.

2.1 Project's pathways towards impact

The project addresses methodology bottlenecks, delivers broadly carried, standardized datasets to compare across protein sources, innovates to overcome barriers, and engages to ensure that stakeholder groups apply the results to deliver mid-term outcomes and long-term impact. To illustrate this, results have been categorized as methodology- (M), dataset- (D), innovation- (I), and engagement- (E) related, and are translated to the intended outcomes and impacts in the tables below. The impact goal of GIANT LEAPS is to substitute the consumption of traditional animal proteins in the European diet by alternative proteins to reach 50% of total protein intake by 2030. Fig. 2.1 illustrates how project results translate via outcome pathways to this ambition and long-term impacts.

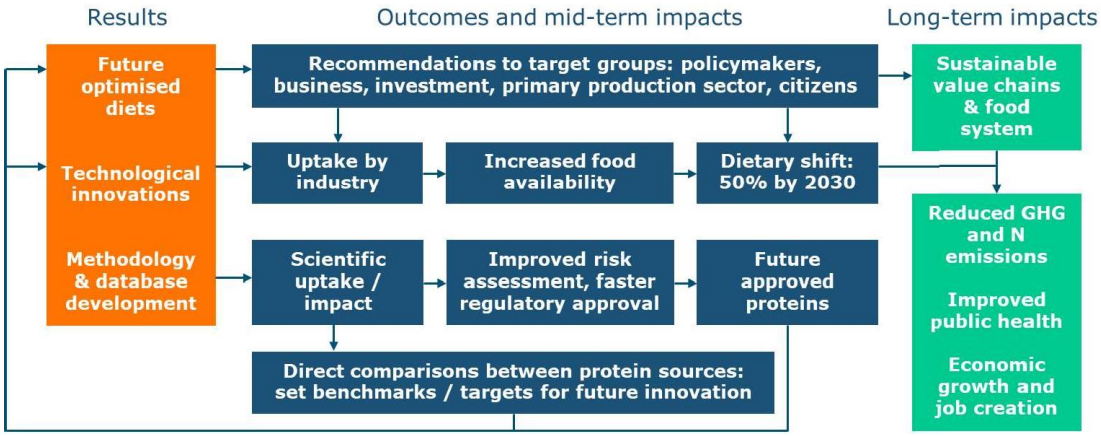

Fig. 2.1: GIANT LEAPS results connected via outcomes to long-term impacts

Specific results related to mid-term outcomes

The role of policymakers, risk assessors and other public governance actors to shape, facilitate and incentivise this complex transition is crucial.[25] The research and innovation (R&I) activities in this proposal provide these key actors with the essential, now-missing, methodologies, datasets, and innovations to take the necessary decisions and actions. Table 2.1 presents the results and knowledge gained from GIANT LEAPS will be used to inform a systemic approach to food policy development and relevant sectoral policies, i.e. policy outcomes.

Table 2.1. Results and outcomes related to food policy development and sectoral policies

GIANT LEAPS results (M-methodology; D-dataset; I-innovation)	**Policy outcomes and KPIs**
Implemented data integration and interoperability platform (SO 6-D) and optimised future diets (SO 7-M)	Guidance on evidence-based policymaking, supported by information and tools, insights into synergies and trade-offs between public health and environmental impacts, and a basis to align relevant sectoral policies on. Updated dietary recommendations based on integrated knowledge on protein quality, co-nutrients and management of risks. Guidance to maximise public acceptance based on insights regarding consumer acceptance data and effects of policy incentives (e.g. taxes, nudging). **KPI:** 1 policy brief on integrated food policy and 2 sectoral policy briefs on public health and consumer education

[24] Food 2030 pathways for action - Research and innovation policy as a driver for sustainable, healthy and inclusive food systems. doi:10.2777/104372I
[25] Science Advice for Policy by European Academies. A sustainable food system for the European Union. 2020

Improved risk assessment methods, case-series and consumer perspectives (SO 3-M), new methods to assess nutritional quality (SO 4-M).	Possibilities to improve the regulatory approval process of novel proteins (e.g. without use of animal studies), identification of critical parameters for food safety monitoring, and recommendations to improve current food labelling policies and to implement an EU post-market surveillance system for monitoring adverse effects. **KPI:** 3 policy briefs on food safety.
Implemented integrated sustainability framework, filled Life Cycle Inventory (LCI) database and combined environmental, economic and social LCAs (SO5-D).	Comprehensive understanding of the environmental footprint and sustainability of alternative proteins. Recommendations to (dis)incentivise production of alternative and/or traditional proteins (land-based and aquatic). Guidance on subsidies and other policy measures, including climate and environmental policies, fair redistribution of profits and fair access to healthy food. Recommendations on climate adaptation strategies for the selected alternative proteins and in general for the protein sector. Inform EU climate policies and strategies, contributing to EU emissions mitigation and climate adaptation goals. **KPI:** 1 policy brief on agriculture and aquaculture sectoral policies, 1 policy brief on environmental policy, 1 for circularity and 1 for EU climate policies and targets.

The two key impact areas in this project are climate-related, including GHGs and biodiversity, and health-related, through more nutritious and healthier food choices. Both **environmental and health impacts of alternative protein sources** are proportional to their level of consumption, so the acceleration of adoption is key to achieving impact. The project delivers results and defines key actions to ensure maximal uptake of results towards a food system change and long-term impact. Two key strategies to reaching mid-term outcomes (described in Table 2.2) are defined:

1. Build **consensus on the optimal selection** of diverse proteins and maximise the use of available alternative protein sources in future EU diets, considering both health and environmental impacts.
2. **Enlarge the number of future available alternative protein sources** that can be applied in foods to enable continuing improvements to diets and their impacts into the future.

Table 2.2. Results and outcomes contributing to optimized protein source substitution in future EU diets

GIANT LEAPS results (M-methodology; D-dataset; I-innovation)	**Outcomes and strategies**	**KPI**	**Target groups**
I– Food prototypes and technological innovations	Maximised use of alternative protein sources in foods and diets	Deliver 6 prototypes and innovations using diverse alternative protein sources	Food industry, retail, out of home
M– Improved food allergy-related consumer risk communication and post-market surveillance strategies	Maximised safe use of alternative protein sources in foods and diets	Report on cross-reactive allergen labelling and post-market monitoring of food allergies	Policymakers, patient and civil society organisations, citizens
M – "Safety by design" toolbox including improved methods for risk assessment	Increased number of available alternative proteins in the future by speeding up and improving risk assessments	Scientific papers describing toolbox and improved predictive in silico and in vitro predictive methods	Scientists, Food industry, policymakers, risk assessors
M– Risk assessment scenario case series	Increased number of future alternative proteins available by clarifying pathways, improved methods and necessary data	Pathways and necessary data submitted for deliberation with panel experts and scientific networks hosted by EFSA	Scientists, Food industry, policymakers, risk assessors
M - Improved in vitro models for assessing protein digestion and bioavailability	Optimised selection of alternative protein sources and blends to meet protein requirements and optimise product design	Models validated with human study and applied to 9 alternative protein sources	Scientists, Food industry
D - Nutritional passports for protein sources and formulated foods	Optimised selection of alternative protein sources, considering co-nutrients, ANF and protein requirements.	Passports delivered for 9 protein sources	Scientists, Food industry
M - Integrated sustainability framework	Optimal selection and use (through processing and optimising side streams) of alternative proteins to fight climate change, halt biodiversity loss, improve ecosystem services and promote circularity	Integrated sustainability framework published in scientific journal. Sustainability profile for 9 alternative protein value chains.	Scientists, primary producers, food industry, policymakers, consumers

D - Filled Life Cycle Inventory (LCI) database	Identified data gaps for sustainability analysis of alternative protein sources	LCI database with existing and project-generated data for traditional and 9 alternative proteins	Scientists, Food industry, policymakers
M - Modelling framework to calculate the impact of diets on health and environmental parameters	Optimised future diets in which alternative protein sources substitute traditional animal proteins	Scientific paper describing framework setup	Scientists, Food industry, policymakers
D - Filled data integration and interoperability platform	Identified data gaps for integrated analysis of alternative proteins across sustainability and health domains	Data platform filled with existing and project-generated data for 28 protein sources	Scientists, Food industry

2.1.2 Expected wider mid- and long-term destination impacts

The GIANT LEAPS project results include evidence and solutions for transforming food systems for health, sustainability and inclusion while maximising co-benefits with respect to the four priorities of the Food 2030 initiative: nutrition and health, climate and environmental sustainability, circularity and resource efficiency, and innovation and empowering communities. These results lead to mid-term outcomes and long-term scientific, economic/technological and societal impacts. The foreseen impacts are provided in Table 2.3.

Table 2.3. GIANT LEAPS scientific, economic/technological and societal impacts

Impact type	Impact	Quantification and specific target groups (if applicable)
Scientific	Standardise and validate methods for determination of protein quality; implementation of improved methodologies and databases accessible to the scientific community; knowledge transfer to researchers in food industry.	4 new/improved methodologies, 2 open-access databases, knowledge transfer to 4 large industry partners.
Scientific	Deliver new, high-quality knowledge to the scientific community.	70 scientific publications, 70 conference presentations,
Scientific	Educate a new generation of scientists in a multi-disciplinary setting and holistic approach to improving the food system.	17 PhDs 15 Postdocs
Scientific / societal	Reduce animal testing for allergenicity and toxicological risk assessments and analysis of digestibility of proteins	Methods leading to up to 30% reduction of animal testing in 2030 presented to and considered by panel experts and scientific networks hosted by EFSA
Economic	Increase return on investment and improved competitiveness through guidance to investors, the primary production sector and food industry in Europe on alternative protein sources to develop / invest in and setting benchmarks for future novel protein sources. The utilization of alternative protein sources requires development of new value chains, providing new job opportunities and a pathway to fairer profit distribution.	Dissemination of results to 40 private sector actors (SMEs, large industries, sector associations) across the value chain through the stakeholder board and dissemination activities
Technological	Increase availability of alternative protein-containing ingredients and foods through uptake of recommendations, food prototypes and technological innovations.	Uptake of innovations by retail, out-of-home and food industry, increasing alternative proteins in food supply up to 30% in 2030.
Societal	Accelerate substitution of animal proteins by alternative proteins in the diets of consumers living in Europe.	50% alternative protein intake (of total) by 2030.
Societal - health	Increase consumer trust through reliable and clear science-based communication about sustainability and health impacts of traditional and alternative protein foods; empower communities through regionally adapted recommendations for sustainable and healthy food choices.	Optimised future diets communicated to 15000 citizens across 4 EU regions by the end of the project.
Societal - health	Improve overall health through a 25% reduction in the consumption of total animal proteins via replacement by alternative proteins by 2030	Estimated[3] years of life saved: 6,215,000

Societal - environmental	Increase the climate resilience of the EU protein sector by delivering a climate vulnerability analysis and adaptation strategies for alternative protein production, making it less vulnerable to events or trends related to climate change.	Increased EU food security and stability for farmers, producers and markets.
Societal - environmental	Mitigate the impacts of climate change, CO_2 emissions and loss in biodiversity through the dietary shift in 2030. Extend the consumer base through a more sustainable, alternative protein-based product portfolio, which contributes to GHG-emission reductions via consumer sensibilization towards sustainability, circularity and carbon neutrality.	Estimated[26] reduction of EU GHG emissions by 2030: 326 MtCO2-eq Estimated reduced EU nitrogen application by 2030: 4901 GgN Increased reuse of nutrients alternative proteins by 15-20%

Other impacts

GIANT LEAPS will directly contribute to the **four Farm to Fork goals of reducing climate footprint, enabling a global transition, creating new opportunities through innovation, and increasing resilience by taking climate adaptation strategies into account**. Indirectly, GIANT LEAPS will also contribute to three of the four Farm to Fork targets: reduction of pesticides (through optimized protein/crop selection), reduction of antimicrobials (through reducing animal protein consumption), and reduction of nutrient losses through improving circularity. By making the **food system healthy, sustainable and more resilient**, GIANT LEAPS contributes to long-term food security and many of the UN Sustainable Development Goals including (3) Good Health and Well-being, (9) Industry, Innovation and Infrastructure, (12) Responsible consumption and Production and (13) Climate Change. More resilient local food supply chains and healthier diets are also essential to avoid COVID-like future shocks to the food system.

2.1.3 Potential barriers to reach impact

The main potential barriers to GIANT LEAPS and strategies to overcome these are:

- Regulatory approval for novel (processed) proteins is a *sine qua non* for implementing alternative protein sources in foods and diets in the EU and, therefore, to reach impact. The activities in WP3 aim to develop improved methods and tools to gather the necessary data while maintaining optimal risk and safety assessment.
- Insufficient industry uptake of the GIANT LEAPS recommendations and innovations hampers the availability of sufficient variety and quantity of foods to consumers to reach impact. Co-creation with industry, utilizationof the extensive networks available to the consortium (B2F network, EFF network, SB) for exploitation, and inclusion of economic aspects in the sustainability analysis are applied to minimise this risk.
- Consumer acceptance is another necessity for reaching impact through the sustainable consumption of alternative protein-based foods as part of daily meals. Insights from quantitative research on consumer acceptance will be used as starting points to define the future optimised diets across 4 regions in Europe. Results of the optimised diets will be used for feedback to consumers to investigate how the dietary shift can be maximised.
- Policymakers need to recognize the credibility and relevance of the optimised future diets as a basis to develop or adapt integrated food policy and relevant sectoral policies, thereby providing incentives for all actors in the food system to enable the transition. GIANT leaps dissemination activities using policy briefs and interactive sessions to explain the concepts and results will be deployed to maximise policy impact.
- Unforeseen impact on primary production of alternative protein sources may influence availability and make the optimised future diets unattainable. However, the applied modelling approach is flexible to be rerun in the future with new data, parameters and proteins to mitigate this potential risk.
- The shift away from traditional animal proteins may disrupt major parts of the food system economy. GIANT LEAPS addresses this risk by proposing a gradual transition, by including the economic sustainability of proteins in the integrated LCA analysis, and by providing new economic opportunities for actors across the value chain related to the utilization of alternative protein sources.

2.1.4 Scale and significance

Around 60-70% of total protein intake is derived from animal sources in European diets.[27,28] The ambition of the GIANT LEAPS consortium is to achieve 50% of total protein intake derived from alternative protein sources, representing a decrease in absolute animal protein intake of 20-30% under the assumption that increased alternative protein intake fully substitutes traditional animal protein intake. Springmann and co-authors published a modelling approach to estimating health and environmental impacts of sustainable diet strategies.[29] Based on their work, a 25% decrease of animal protein intake would reduce GHG emissions in 2030 by 326 MtCO2-eq, reduce EU nitrogen application by 2030 of 4901 GgN and save 6,215,000 life-years through long-term health impacts.

[26] Based on Springmann et al. (2018) Lancet Planetary Health 2;10, pp e451-e461. doi.org/10.1016/S2542-5196(18)30206-7
[27] doi: 10.1038/ejcn.2009.73
[28] doi:10.3945/jn.113.177113
[29] doi.org/10.1016/S2542-5196(18)30206-7

2.2 Measures to maximise impact - Dissemination, exploitation and communication

2.2.1 Dissemination, exploitation and communication

All partners in GIANT LEAPS are committed to deliver on the objectives and create widespread impact from the project results. The combined experience, expertise and wide network of our consortium members will ensure that we reach maximum impact with all key actors identified. A fully integrated plan for communication, dissemination and exploitation (Task 8.1) involving all partners will be developed and implemented within 6 months after the project starts and updated continuously throughout the project. Its objective is to ensure that the impact of the project is **immediate and continues beyond the life** of the project, addressing all relevant target groups.

GIANT LEAPS will disseminate and promote the results of the project to key stakeholders active and related to the food supply chain from farm to fork, e.g. **primary producers, food and ingredients producers, food processors and distributors, wholesalers, retailers, food service, restaurants, consumers, opinion leaders/regulators, the scientific community, food manufacturers' associations, media, consumer organisations and the public**. The geographic coverage ensured by the Consortium members, who have networks across Europe, will enable a far-reaching and systemic dissemination. Each GIANT LEAPS partner will be actively involved in the project's dissemination activities, which will enhance the sustainability of the project results. They will also assist in translating any dissemination and communication materials into the local languages and attend local events.

GIANT LEAPS will focus particularly on **dissemination of results to policymakers** via dedicated policy briefs. The project will contact policymakers and risk assessors at the EC including EFSA, at the WHO/FAO, and in Member States of the EU. Contacts will be established through in-person and online meetings to identify the views of policymakers on issues relevant to alternative proteins, dietary shift, environment and health, as well as via a workshop. Combined with the project approach and results this information will be used to generate policy briefs on 9 key topics of the project. Policy briefs are a unique means of tailored communication and will contain adequate information, fully supported by science and presented in tangible format for policymakers to digest.[30] They will be drafted within WP8 in liaison with the other work packages in the project on the following topics:

- 1 policy brief on integrated food policy (dietary goals 2030, interrelated goals & impacts for sectoral policies)
- 2 sectoral policy briefs on public health and consumer education
- 3 policy briefs on food safety
- 1 policy brief to inform both agriculture and aquaculture sectoral policies
- 2 policy briefs on environmental policy and circularity

Collaboration with other food-related projects and the use of national contacts: GIANT LEAPS will multiply its outreach through joint dissemination efforts with other projects and initiatives, including but not limited to the four projects funded under topic *LC-SFS-17-2019: Alternative proteins for food and feed* and *HORIZON-CL6-2021-FARM2FORK-01-02: Developing sustainable and competitive land-based protein crop systems and value chains*. This will result in joint social media campaigns, joint events, participation at each other's events and workshops, and other joint dissemination and communication actions. The outreach and contribution of the GIANT LEAPS consortium and regional/national networks and initiatives will be crucial in engaging all stakeholders for the proper dissemination, communication and further uptake of our results and best practices.

A comprehensive communication strategy will be deployed. The project **website** will be the primary interface for communicating the GIANT LEAPS results, featuring links to the partner websites, project updates, news articles, events, social media widgets and contact information. An electronic **newsletter** will be produced every 6 months, with recent highlights from the project. Several project **leaflets** will be developed for distribution to partner networks and at conferences, exhibitions and other events in addition to a project media kit. The project will have a **social media presence** through Twitter, LinkedIn and ResearchGate, to ensure wider dissemination to many target audiences. A **LinkedIn page** will be created for the GIANT LEAPS community to exchange information about project developments and gather views on specific aspects of the project. Every effort will be made to publish the work of the consortium via the **media, publications, conference presentations and workshops**, as well as through the Commission and industry bodies. A **final GIANT LEAPS conference** will be organised to raise awareness about the dietary shift and the role alternative proteins can play in this, to share the results of the Giant Leaps project, but also to give floor to related other initiatives in society. Additionally, the Annual EFFoST Conference will dedicate a special session for presentations from the GIANT LEAPS partners. An **e-learning platform** will be set-up with courses for multiple target groups. The objectives will be to (i) educate school children and students, (ii) to provide new knowledge and upgrade practical skills of professionals, (iii) raise awareness in society, and (iv) contribute to overcoming people's biases towards alternative proteins and the dietary shift. Results of the project will be disseminated via **reports, scientific papers, policy briefs and technical articles**. All public communication will be made open access, to facilitate scientific exchange. Open publications will also be shared through the personal profiles of the authors on ResearchGate. Table 2.4 summarises the main dissemination tools and channels used to reach the various audiences.

[30] Good examples are Policy Briefs as practised by the WHO (examples: LINK1, LINK2, LINK3)

Table 2.4. GIANT LEAPS Dissemination and communication tools

Dissemination tool	Target audience			KPI	Means of verification
	Scientific community	Industry & SMEs	Public at large		
Project website	Y	Y	Y	10000 visitors + 25000 page views. 30% spending more than 1 minute on the website.	Web traffic statistics collected and documented in the activity reports
Project materials (leaflets, brochures, A/V publications)	Y	Y	Y	3 videos with over 750 online views per video; 1000 project leaflets distributed	Tracking spreadsheet: input from partners and YouTube views
Newsletters	Y	Y	Y	7-8 newsletters share with our existing mailing lists	Newsletters distributed and published via the project website and shared with our partners' mailing lists
Scientific publications	Y	Y		6-10 conference papers and journal articles	Activity reports
Participation at scientific conferences, fairs and events	Y	Y	Y	40+ attended 20+ talks	Activity report and tracking spreadsheet: input from partners
Civil society and consumer engagement	Y	Y	Y	1 event/country, 2 European food events (fairs, festivals), 500+ citizens engaged	Activity report and tracking spreadsheet: input from partners
General audience articles/media		Y	Y	10+ non-scientific journal news	Activity report, press releases
Social media (Twitter, LinkedIn, ResearchGate)	Y	Y	Y	150+ project-specific tweets via consortium partners 500+ members on GIANT LEAPS LinkedIn group page	Twitter/LinkedIn analytics and statistical reports
Radio/television broadcasting	Y	Y	Y	At least 1 national presence/country	Recorded sessions
e-learning courses	Y	Y	Y	At least 3 modules with 2-3 chapters each; with 10-15 min per chapter	Number of completed chapters followed in the LMS
Policy briefs/fact sheets	Y	Y		9 Policy briefs and/or fact sheets distributed to >100 policymakers	Tracking spreadsheet: input from partners
Press kits	Y	Y	Y	25 media kits downloaded	Activity report
Press conferences and press releases	Y	Y	Y	4+ press releases	Activity report
GIANT LEAPS final conference	Y	Y	Y	> 120 participants > 7 speakers	Online social media and website communications. Registration list.

Stakeholder and international network engagement

The GIANT LEAPS Stakeholder Board (SB) is an open group that will be further extended before, during and after the project. It is composed of representatives from all kinds of organisations related the supply chain from farm to fork, e.g., primary producers, food and ingredients producers, food processors and distributors, wholesalers, retailers, food service, restaurants, and consumers, opinion leaders/regulators, the scientific community, food manufacturers' associations, media, consumer organisations and the public. In addition, important public sector stakeholders such as WHO and EFSA will be contacted to join after the project is granted. The SB will be part of the activities in WP1 and WP8 and has both a consultative role on the project content and approach, and a role to maximise the dissemination and exploitation of the project results. The current participants are listed in Table 2.5.

Table 2.5. Participants of the GIANT LEAPS Stakeholder Board

Protein & flavour suppliers	Fuji Oil; Firmenich; Cosun; Cargill; Avebe; Quinoa Marche
Food & dairy manufacturers	Valio; CJ; Fazer; Hochland / Simply V; Westland; Lantmännen; Savencia / Soredab; Oatly; Nestlé; Kellogg*
Technology suppliers	SPX Flow; CER Groupe
Economic / investment	World Economic Forum*; CounterFactual Ventures
Boards & platforms	European Alliance for Plant-based Foods; Good Food Institute; European Federation of the Associations of Dietitians; International Platform of Insects for Food and Feed
Public sector / food authorities	FAO Office of Climate Change, Biodiversity and Environment*; former EFSA panel members: prof I. Tetens, prof A. Flynn
Research institutes	ILVO Vlaanderen*; Institute of Food Research and Product Development, Kasetsart University, Thailand

*Note: *formal confirmation pending; Letter of Support received from all other participants*

Large international networks will be leveraged for communication and dissemination. EFF is the European affiliate of the International Union of Food Science and Technology and connected with more than 130 societies, institutes and universities across Europe. EFF also collaborates with organisations such as the European Food Information Council (EUFIC), the International Life Sciences Institute Europe (ILSI Europe), the Global Harmonisation Initiative (GHI) and the National Food Technology Platforms (NFTPs). Moreover, B2F has a worldwide network of 500+ small and large businesses and other organisations. Combined with all individual partners, including 5 large industries, the GIANT LEAPS consortium has a tremendous network and potential for successful dissemination and impact.

At the end of the Giant Leaps project, a **final report** will highlight the dissemination achievements according to the activities outlined in the communication/dissemination plan. The report will also include a 2-year plan for disseminating the results beyond the end of the project, including various activities such as scientific publications, conferences, networking, and follow-up projects. The success of the dissemination plan will be measured by collecting the feedback and comments, press coverage, queries posed by stakeholders, comments articulated by participants at workshops and seminars, and detailed website access statistics on a regular basis.

Exploitation

The exploitation routes will be investigated in WP2-5 (scientific research) and WP8 (policy recommendations). The involved partners are experts and referents in fields of scientific communication, policy support, technology transfer as well as business development for the food industry (see partner profiles). EFF, in alignment with the WR Innovation and IPR (IIPR) Manager and with support of the members of the consortium, will act as coordinator of the exploitation activities and ensure that all partners as well as potential and confirmed exploitation partners will work towards a successful technology and knowledge transfer to ensure efficient market uptake. EFF will be responsible for establishing **targeted Dissemination and Exploitation strategies of Project Results**. The entire project consortium will discuss the strategy and the corresponding document (PEDR) will be delivered at Month 6 and updated annually for the remaining project duration. The first phase of exploitation activities will aim to make detailed exploitation plans for each partner. The second phase aims to raise awareness of the project among target users and ensure engagementof relevant stakeholders. The third phase aims to communicate the developed products as well as provide long-term exploitation support. An overview of the main exploitable assets is shown in Table 2.6.

Table 2.6 GIANT LEAPS Exploitable assets

Exploitable asset	Unique value proposition (UVP)	Exploitation type & tentative BM	End-users	Partner(s) involved
GIANT LEAPS stakeholder board Platform	A sustainable GIANT LEAPS platform with stakeholders who exchange experiences and knowledge on the production of alternative protein foods, the dietary shift and policy measures	Commercial Product B2B	Stakeholders, including food industry, associations, NGOs, public authorities, policymakers at EU and national level	B2F EFFoST FSN
GIANT LEAPS portfolio of demand-driven alternative protein solutions (WP2)	Processing technologies for developing sustainable, healthy and delicious alternative protein-based foods to obtain a diverse group of foods made of alternative protein ingredients	Commercial Product B2B/B2C depending on the interest	Food producers, ingredient producers, food processors,	Partners in WP2

GIANT LEAPS dairy and meat alternatives food models	Production of dairy and meat alternative food models with superior functionality than animal alternatives	Commercial Product B2B/B2C depending on the interest	Food producers, ingredient producers, food processors,	Partners in WP2
GIANT LEAPS Pilot scale production	Pilot scale production of functionalised selected animal alternative products	Commercial Product B2B/B2C depending on the interest	Food producers, ingredient producers, food processors,	Partners in WP2
Education programme	Stimulative, interactive and innovative educational program aimed at raising awareness about the role of alternative proteins in the dietary shift	Commercial Service provision, subscription-based or selling educational content	Scholars and students Food professionals General public	EFFoST B2F EM

EFF will support the development of business plans considering any exploitable results of each partner within the consortium. In addition to potential benefits for industrial partners, also academic partners will be positively affected by outcomes, due to the high innovativeness of the proposed solutions and their societal and scientific relevance. Coherently with the type of actions implemented in the GIANT LEAPS work plan, exploitable results will be valorised in the short-term impact (<3 years of project implementation) and/or in the long-term impact (4-10 years) by project partners and by external end-users, who will be preliminary identified through stakeholder analysis.

All technologies, processes, services and products developed by the partners will be considered for exploitation. Business models to strategically manage marketability beyond the duration of the project will be developed. The plan will also include a roadmap of action to reach TRL9. The demand side in general and consumer concern in particular will be studied through consumer acceptance studies and stakeholder workshops in WP1. Outcomes of these activities will be used to develop recommendations on how to communicate about new alternative protein-containing products to consumers, to inform and alleviate concerns, reduce misinformation and minimise the likelihood of rejection.

2.2.2 Intellectual property management

The consortium will seek to make best use of the exploitable results of the project, with special focus on seeking IPR protection for innovation-type results. For new or improved methodologies, technologies and datasets, the aim is to pursue open access and open science practices (see also section 1.2 Open Science & Data Management). When considering IPR, the following should be emphasized for consolidating sustainable value creation in the EU:

- The high relevance of this project lies in the fact that "alternative protein sources" solutions are provided, considering positive economic and social impacts in addition to the environmental aspects, for **widespread use across the food sector**, along the entire food chain, and not limited to the EU alone.
- The high technical quality of the project is associated with innovative technologies with large potential to valorise at the end of the project, requiring **sufficient freedom-to-operate** by the exploitation partners.

The partners acknowledge the strategic importance and sensitive nature of back- and foreground knowledge provided by any other partner, and that disclosure of such information will be subject to specific obligations concerning confidentiality and use. A Consortium Agreement will be signed before the start of the project which details relations and regulations regarding IPR between the partners. This will enable all partners to carry out their work on the project and/or exploit the results obtained in accordance with the Grant Agreement. IPR guidelines (WP9) will be developed by the IPR manager at the beginning of the project, distributed to and agreed upon by the consortium partners.

A system for monitoring the results generated during the implementation of the project is created to have a common repository shared between the partners, serving as a collective knowledge management and exchange system. Partners and WP leaders periodically send preliminary results to the Executive Committee (ExC) for evaluation of possible exploitation or dissemination, including communication between the partners, externally with the SB, other dissemination and/or communication of the activities carried out, through the publication of scientific articles, etc. The ExC will always respect the appropriate protection of IP rights in the various dissemination-exploitation routes, and the information will be shared with all partners in the reserved area of the project portal/collaboration site.

2.3 Summary

Specific Needs	Expected Results	D & E & C measures
The protein transition to a healthy and sustainable food system is needed. Technological, nutritional and sensory bottlenecks hamper implementation of sustainable protein sources in food products, and require **innovations** to be overcome. An integrated framework is urgently needed to enable comparisons between protein sources taking all relevant domains into account (climate, environmental impact, safety and health). This is prohibited by gaps in single domain predictive **methodologies** (toxicology, allergenicity, digestibility) and the lack of comprehensive comparative **datasets**.	**All missing elements of an integrated framework to evaluate protein sources** are developed to enable the shift towards a sustainable food system and healthy nutrition: - **innovative prototypes** that unlock the potential of sustainable protein sources in foods, products and diets. - predictive in silico and in vitro **methodologies** for safety and health parameters. - **datasets** and cloud platform for data integration, analysis and comparison. - models to estimate and optimize the environmental and health impacts of future diets and the shift towards alternative protein foods.	**Dissemination towards policymakers**: Policy briefs with recommendations for integrated food policy development and sectoral policies are generated, in addition to a final conference to present the project results and engage with stakeholders. Specific partners and Stakeholder Board participants (e.g. World Economic Forum, large industries, (ex-)EFSA panel members) will support to maximise impact. **Dissemination towards the scientific and R&D community**: High-impact scientific publications and presentations at scientific conferences. A training program is developed for interested parties on the project's outcomes. Contributions to Fairs and to branch-related journals to facilitate industrial uptake. **Communication towards citizens:** Major project developments will be communicated through the project website, and through press releases, articles in local newspapers, blog posts, infographics, videos, etc. Outreach potential will be maximised by using social media including Twitter, LinkedIn and ResearchGate. **Exploitation:** Database and cloud platform will be developed to be maximally open-access and interoperable. Food prototype and technical innovations will be patented if possible and exploitation options will be pursued within and outside of the consortium's wide industry, entrepreneurial and scientific networks.

Target groups	Outcomes	Impacts
Public sector: National/international bodies, regulators and legislators, public health authorities. **End-users**: Consumers & consumer associations, patient organizations and dieticians. **Agri-food sector**: Entire value chain including ingredient and food processing & manufacturing, primary production sector, catering and retail companies. **Scientific community** related to food systems, sustainability, nutrition, health, safety, allergenicity and food technology.	**Policymakers, businesses and investors** use the integrated assessment and recommendations to guide long-term policy (integrated and sectoral) and investment strategy regarding production and consumption of alternative proteins **Scientists** in academia and industry utilize improved in vitro models to predict allergenicity and digestibility tests for new protein sources and products. Integrated assessment methodology remains in use after the project and gets updated in future. **SMEs and large food industries** exploit new technologies, innovative food prototypes, datasets and future diet insights to bring new alternative protein products to market. **Citizens** will be empowered with regionally-adapted recommendations to make informed, evidence-based food choices and actively contribute to positive health and environmental impact, increasing public trust as a co-benefit. A wider range of options of alternative proteins and products will be available to **manufacturers and consumers**.	**Scientific:** Advanced integrated assessment and comparison of impacts of alternative and traditional protein sources on climate, biodiversity, sustainability, circularity, health and safety parameters. Advancement of *in silico* modelling and *in vitro* predictive models related to health and safety. Training of a new generation of researchers. **Economic/Technological:** Economic and job opportunities, and reduced risk and expense, related to the development and use of alternative proteins in relation to start-up companies and large industry. Increased availability of alternative protein sources **Societal:** Dietary shift to 50% alternative proteins in 2030 and diversification of protein sources for human nutrition. Halting biodiversity loss, improved ecosystem services, lower ecological footprint, improved food system circularity and resilience. Increased population health, reductions in animal testing and improved societal acceptance of alternative protein sources.

3 QUALITY AND EFFICIENCY OF THE IMPLEMENTATION`

3.1 Work plan and resources

Summary of the overall structure of the workplan

The multi-disciplinary and interrelated research activities of GIANT LEAPS are grouped in Work-Packages (WPs) for the main themes of the dietary shift, representing the most logical and optimal work division. Fig. 3.1 illustrates the main interrelationships and flows of information between the nine individual WPs. Each WP consists of 3 to 6 manageable Tasks to reach the objectives of the respective WP. The focus of the WPs is summarized in Table 3.1.

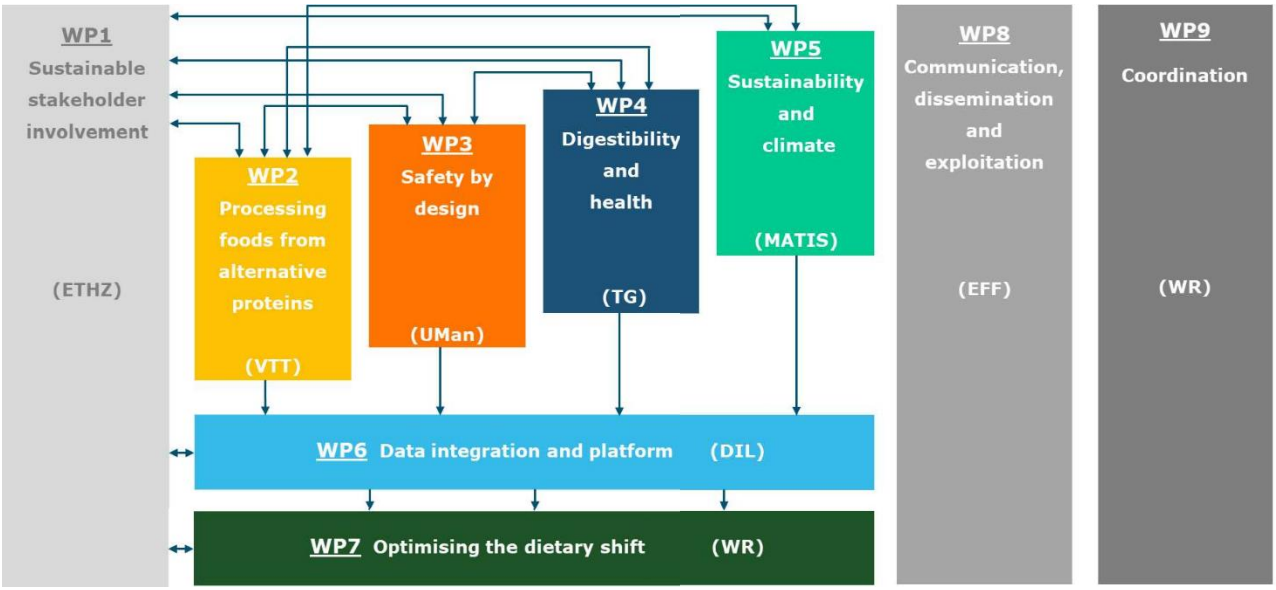

Fig. 3.1: Work Package structure and main interrelationships and information flows (Pert chart)

Table 3.1. Summary of the focus of the 9 Work Packages

WP	Main focus of the GIANT LEAPS Work Packages
1	Engagement with relevant stakeholders, such as European and national policymakers, food companies, NGOs and consumers, to retrieve their views and feed their information into the project, and vice versa. Consumer studies on barriers and drivers for acceptance of alternative proteins in daily diets, as well as consumer trade-offs between different aspects of the dietary shift.
2	Investigating processing technologies for developing sustainable, healthy and delicious foods from alternative proteins, as alternatives for egg, meat and milk protein-based foods
3	Develops a toolbox that enables the safety by design approach across a range of hazards for alternative proteins (including nutritional imbalance), demonstrated for several case studies, and addresses critical issues in allergenicity assessment of alternative proteins
4	Assesses the digestibility of alternative proteins foods, both *in vitro* and *in vivo*, and as part of that improves and validates existing *in vitro* digestion models and holds a human trial.
5	Assessing the sustainability (from environmental, economic and social impacts), biodiversity and climate change mitigation/adaptation potential from the production of several alternative proteins.
6	Set ups a data integration platform assuring the interoperability and data access between GIANT LEAPS partners, and beyond. The platform will be based on a database management system hosted on a cloud server and connected through defined ontologies and data protocols
7	Estimates and optimizes the impacts of the foreseen and consumer accepted dietary shift on the environment and human health, in comparison with traditional animal protein diets, with the ultimate aim to provide the scientific knowledge based to all stakeholders.
8	Maximizes the outreach and impact of the GIANT LEAPS products and innovations in terms of enhanced market and business opportunities, growth and jobs in Europe, as well as the foreseen dietary shift, and dissemination of non-proprietary results.
9	Coordinates the GIANT LEAPS project and consortium at the 3 levels of operational, strategic and organisational levels, and ensures proper Ethics and IPR management.

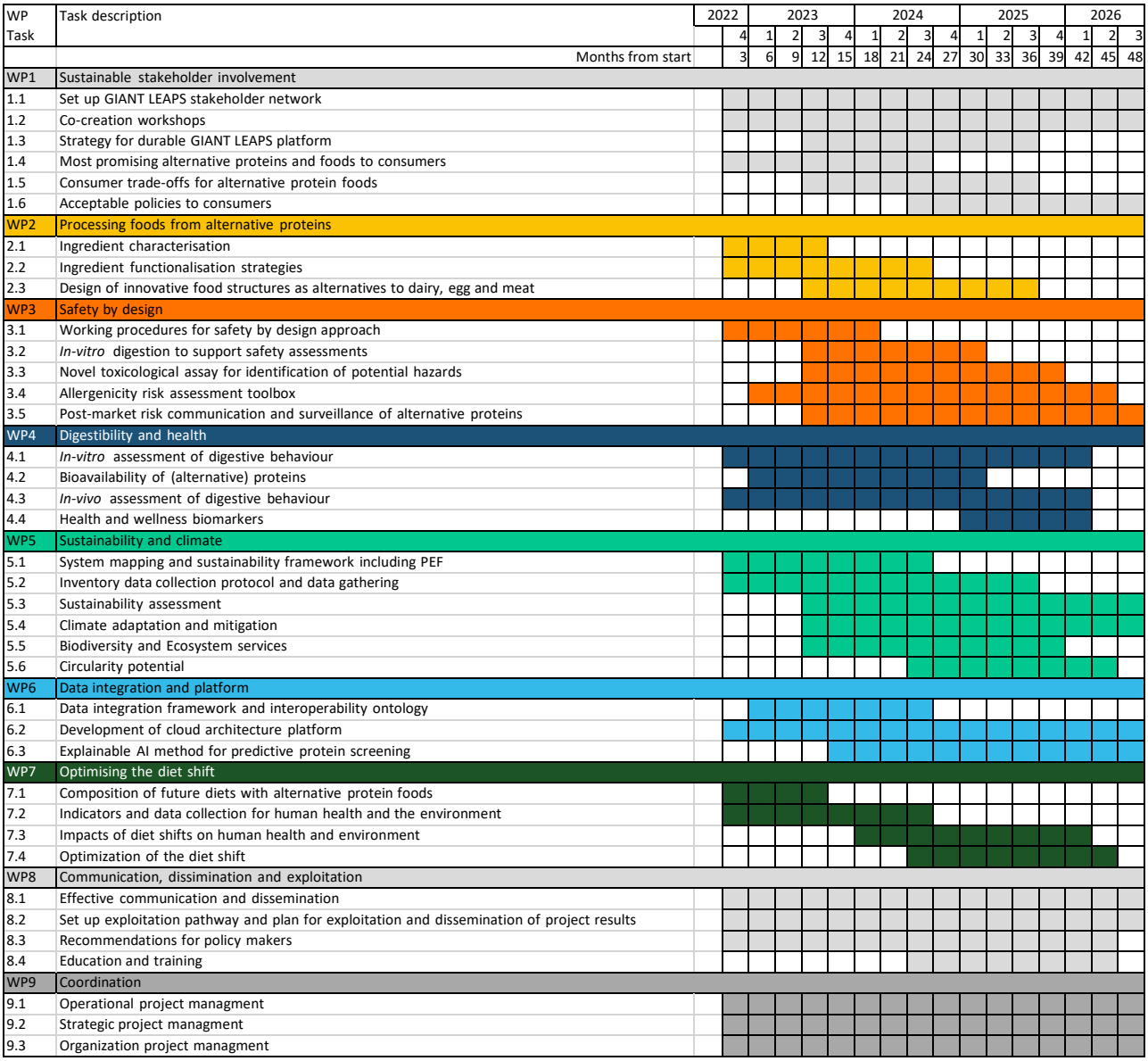

Fig. 3.2: GIANT LEAPS Gantt chart

Table 3.2a: List of work packages

Work package No	Work Package Title	Lead Participant No	Lead Participant Short Name	Person-Months	Start Month	End month
1	Sustainable stakeholder involvement	33	ETHZ		1	48
2	Processing foods from alternative proteins	5	VTT		1	36
3	Safety by design	6	UMan		1	48
4	Digestibility and health	11	TG		1	42
5	Sustainability and climate	3	MATIS		1	48
6	Data integration platform	10	DIL		1	48
7	Optimising the dietary shift	1	WR		1	45
8	Communication, dissemination, exploitation	17	EFF		1	48
9	Coordination	1	WR		1	48
			Total person-months:			

Table 3.2b: Work package descriptions for each work package

Work package No.	1				**Lead beneficiary**				ETHZ		
Work package title	Sustainable stakeholder involvement										
Participant No	1	2	3	4	5	17	18	23	24	27	33
Participant	WR	IRTA	MATIS	UCC	VTT	EFF	B2F	FSN	Mosa	CAP	ETHZ
Person months:											
Start month	1						**End**		48		

Objectives

O1.1 Identify the relevant stakeholders and to interact with them in co-creation workshops
O1.2 Set up a strategy for a sustainable GIANT LEAPS stakeholder platform
O1.3 Identify consumer acceptance of alternative proteins, and potential barriers for acceptance
O1.4 Identify policies acceptable to consumers to achieve a shift in replacing animal protein foods

Description of work

Task 1.1 GIANT LEAPS stakeholder and scientific network. Duration: M1-48 (**B2F**, WR, IRTA, MATIS, UCC, EFF, FSN, Mosa)

Stakeholders, including food industry, associations, NGOs, public authorities, and policymakers at European and national levels, for the GIANT LEAPS project will be identified. Starting point is the project's (open) Stakeholder Board (SB) which already involves >25 different stakeholders (see Table 2.5). Additional stakeholders will be identified and approached to join the SB to extend it to a least 50 stakeholders halfway through the project. Protein category Theme Leaders (WR, IRTA, MATIS, UCC, Mosa) will connect with running *LC-SFS-17-2019* projects and align all WP activities on the respective protein categories. Connection to other projects including *HORIZON-CL6- 2021-FARM2FORK-01-02*.

Task 1.2 Co-creation workshops with stakeholders. Duration: M1-48 (**B2F**, EFF, FSN)

Co-creation workshops (three) with all stakeholder groups and consortium partners will be organized. These will be held (live) in conjunction with three of the project's annual consortium meetings (kick-off, midterm and final). In the first two workshops, the past and future will be scoped; the research questions and methods of the project will be discussed in-depth; and input, ideas and directions will be sourced from each of the various stakeholder groups. The gathered input will then be brought into the consortium meetings for evaluation of possible uptake in the project activities. Also, specific input needed in other WPs will be collected from the stakeholders, such as for WP2, 7 and 8. The last workshop will focus on the integration of project results into stakeholder's strategies.

Task 1.3 Strategy for sustainable GIANT LEAPS platform development. Duration: M12-36 (**B2F**, EFF, FSN)

A strategy will be developed for extension of the GIANT LEAPS SB both beyond Europe and after the project duration, aiming at achieving a sustainable GIANT LEAPS platform with stakeholders who exchange experiences and knowledge on the production of alternative protein foods and the foreseen dietary shift, and policy to achieve this (B2F, EFF, FSN) . Setting up the platform in *other continents* and *continuation after the project* (B2F) will benefit the worldwide production, uptake and consumption of alternative protein-based foods.

Task 1.4 Promising alternative proteins and foods to consumers. Duration: M1-24 (**ETHZ**, VTT, CAP)

Internet-surveys will be held aimed to identify what kind of alternative proteins consumers would like to see in certain food product categories, and to identify drivers and barriers for consumers' willingness to accept alternative protein foods as part of their daily diet. The survey will cover both the GIANT LEAPS long- and short-list of alternative proteins. The surveys will be conducted in one country from each of the four regions in Europe considered in WP7 (North, West, East, South), among a total of >1,000 participants. Results of the survey will be used in the selection of alternative proteins in WP2, in defining future diets in WP7, and for selecting interested consumer groups for dissemination in WP8.

Task 1.5 Consumer trade-offs for alternative protein foods. Duration: M12-36 (**ETHZ**, VTT, CAP)

Online surveys among consumers will be held (ETHZ) to determine how important naturalness, healthiness, animal welfare, sustainability, price, nutritional and sensory characteristics are for the acceptance of alternative proteins and derived foods that are identified as the most promising candidates in Task 1.4 (selected set of alternative proteins). For product development, it will be important to understand what trade-offs consumers are willing to make, e.g. regarding product characteristics versus price. Data will be collected in the same countries as for Task 1.4 (VTT, CAP). The questionnaire will be designed in close collaboration with WP2 and WP7. Results will feed back into these two WPs, and in Task 1.2.

Task 1.6 Acceptable policies to consumers. Duration: M24-46 (**ETHZ**, B2F, FSN)

Internet experiments will be conducted among consumers to determine the policy measures (such as labelling, taxes, subsidies, nudging or education) that are most promising for increasing the consumption of alternative proteins and to decrease the consumption of animal proteins. Furthermore, public acceptance of these various policy measures will be assessed. Data will be collected in the four European countries also considered in Task

1.4, differing in cultural habits and food culture. Results will feed into Task 1.2 (3rd workshop) and WP8.

Deliverables

D1.1 Reports (three) with results of the co-creation workshops, B2F (M3, 24, 48)
D1.2 Scientific publication on promising alternative proteins (foods) to consumers, ETHZ (M24)
D1.3 Scientific publication on consumer trade-offs and dietary shift policies, ETHZ (M46)

Work package No.	2			**Lead beneficiary**						VTT		
Work package title	Processing foods from alternative proteins											
Participant No	2	3	4	5	19	20	21	25	26	28	31	34
Participant	IRTA	MATIS	UCC	VTT	VM	SF	GCL	NFB	UL	FC	BD	AGT
Person months:												
Start month	1							**End**		36		

Objectives

O2.1 Characterize "short list" of alternative proteins, and define clean label functionalisation strategies for improved technological, sensory and nutritional functionality in foods (co-work with WP3 and 4)
O2.2 Develop *in situ* functionalisation approaches as part of food design through use of thermo-mechanical and bioprocess technologies
O2.3 Design healthy, sustainable and delicious meat, dairy and egg alternatives through innovative food processing and formulation approaches (co-work with WP1, 3, 4, and 8)
WP2 focus is on the alternative proteins short list (except cultured beef due to low volumes) shown in Table 1.2.

Description of work

Task 2.1 Ingredient characterisation. Duration: M1-12 (**MATIS**, IRTA, UCC, SF, GCL, NFB, BD, AGT)
Available data for each raw material are collected to uncover knowledge gaps and ensure the best use of funds by preventing duplication of analysis. Functional properties of each alternative protein ingredient (i.e. composition, conformation) and application conditions will be studied to understand protein-protein interactions. Additionally, protein techno-functionality aspects such as solubility, emulsification, foaming, gelation, cohesion-adhesion, elasticity, viscosity, fat absorption, wettability, flavour binding and hydrophobic properties will be investigated. Sensory properties of ingredients (taste and flavour) will be evaluated in suspensions. The proximate composition, including amino acid and fat composition will be studied.
Task 2.2. Ingredient functionalisation strategies. Duration: M1-24 (**UCC**, IRTA, VTT, VM, NFB, UL, FC, BD, AGT)
Combinations of different protein sources in different complex food model systems (from liquid to solid) will be optimized by using biotechnological tools, such as enzymatic treatments and microbial fermentation. A comprehensive evaluation of the functional properties of the modified proteins (emulsification stability, water binding etc.) and the evolution of their structure conformation (size distribution, charge etc.) will be performed, as in Task 2.1. The resulting functionalised protein ingredients will be applied in different food/beverage model systems in Task 2.3. Microstructure (using CLSM, SEM) and macrostructure investigation, using a multistage approach such as rheology (oscillation and rotational tests) and tribology analysis (allowing the understanding of the oral processing of food products), and experimental food development will be performed. This will enable developing a powerful prediction tool for the subsequent optimisation of future food prototypes.
T2.3: Design of innovative food structures as alternatives to dairy, egg and meat. Duration: M12-36 (**VTT**, IRTA, MATIS, UCC, VM, UL, FC)
Healthy, sustainable and desired meat (wet and dry extruded (VTT, UL), dairy (drinks, yoghurt type gels, cheese like spreads (UCC, IRTA, FC), culinary products (e.g.: patties, meat balls) (IRTA, MATIS) as well as egg replacers for fine bakery products (UCC) will be developed taking into account ingredient properties (Task 2.1) and functionalisation strategies (from Tasks 2.1 and 2.). Consumer insights obtained in WP1 will guide the selection of ingredients for dedicated food applications. Technological, nutritional and sensory quality of foods will be of utmost importance, where iteration loops between Tasks 2.1 and 2.3, as well as WP3 and 4 will be included, allowing space for improvement throughout the project. Nutritional, structural, sensory attributes of the ingredients will be upgraded (co-work with Task 2.2) via utilising the understanding of the complex component interactions and routes to modulate. Food-compatible and label-friendly technologies for improving the functional properties of protein ingredients will be performed in co-work with Tasks 2.1 and 2.2. The work will include limited hydrolysis by enzymatic means (of proteins or other components such as phytate and cell wall components), flavour modulation by enzymatic means or with tailored solutions, enzymatic protein cross-linking or deamidation, shear-induced processing (e.g. high pressure homogenisation), and fermentation. The functionalisation toolbox is

essential to create synergy between components and take advantage of multi-functional properties of coexisting polymers within the protein ingredients. Most potential functionalisation approaches will be selected based on the knowledge on ingredient performance obtained both in lab-scale-model and complex matrix applications (link to Tasks 2.1 and 2.2). Developed foods will be characterised for rheological, textural properties, microstructure, flavour and mouthfeel. Selected samples will be sent to WP3 and WP4 for further analyses.

Deliverables

D2.1 Report on alternative protein ingredients characterization, MATIS (M12)
D2.2 Report on functionalisation strategies for alternative protein ingredients, UCC (M24)
D2.3 Report on alternative protein ingredient-based innovative food structures, VTT (M36)

Work package No.	3	**Lead beneficiary**										UMan			
Work package title	Safety by design														
Participant No	1	5	6	7	11	12	14	23	24	26	29	30	31	32
Participant	WR	VTT	UMan	UNa	TG	INRAE	MUS	FSN	Mosa	UL	DAAB	ROQ	BD	DNR
Person months:														
Start month	1								**End**		48			

Objectives

O3.1 Develop a "safety by design" approach of alternative future proteins using in silico and in vitro methods
O3.2 Identify compounds of toxicological concern by using in-silico and novel in-vitro methods
O3.2 Integrate in silico and in vitro methods for assessment of allergenic risks related to cross-reactivity and de-novo sensitization
O3.3 Develop effective post-market surveillance approaches to understand how the paradigm shift in dietary proteins may impact on adverse reactions and (particularly) food allergies

Description of work

Task 3.1 Working procedure for safety by design approach. Duration: M1-18 (**UMan**, WR, VTT, UNa, TG, FSN, Mosa, UL, ROQ, BD). A safety data gap analysis for future alternative proteins will be undertaken, including microbiological hazards (including viruses and prions), toxicological (general and genotoxicological) and chemical hazards, nanoparticles, biotoxins, nutritional quality (including antinutritional factors), allergenicity (including any history of sensitisation) and immune modulators. This will be based on existing and upcoming regulation, literature review with data gap analysis, and expertise from project partners and Stakeholder Board members, using exemplar ingredients from at least five alternative proteins ingredient types, of which 2-3 from the short list (e.g. rapeseed, crickets) and 2-3 exploratory sources (e.g. hemp, leaf proteins, algae). The resulting case studies will be used to outline an *in vitro* testing toolbox and associated requirements for novel test methods that embraces a safety by design approach for alternative proteins that takes amount of existing and draft safety regulations. This will provide a focus for WP3 to fine-tune the toxicity testing approach (Tasks 3.2-3.5) and identify a list of alternative proteins on which Tasks 3.2-3.4 will focus. This will also include comparative traditional protein ingredients of high (cow's milk, peanut), moderate (soybean) and low (rice/maize) allergenicity.
Task 3.2 *In-vitro* digestion to support safety assessments. Duration: M12-30 (**UNa**, WR, UMan, INRAE, UL). Digesta based on INFOGEST protocols will be prepared (WR) for Task 3.3. The potential to use untargeted mass spectrometry (MS) analysis to identify bio-accessible novel compounds of digesta will be explored, and *in-silico* QSAR methods applied to identify novel compounds of concern (UMan, UNa, UL). WP4 Infogest digestion protocols will be mapped to those used routinely for allergenicity risk assessment and proteins and peptides in digests analysed using gel-based and MS-based methods (UNa, UMan). Mass distributions and sequence data will enable kinetic analysis to identify persistent fragments (Tasks 3.3 and 3.5) and provide consensus protocols that will be trialled and applied (by WR, UMan, INRAE) in Tasks 3.4 and 3.5.
Task 3.3 Novel toxicological assay for identification of potential hazards. Duration: M12-40 (**WR**, ROQ, BD). The impact of selected alternative protein samples (Task 3.1) and *in vitro* digests prepared using the WP4 protocols (Task 3.2) on cell lines representative of immune cells that are known to form the mononuclear phagocyte system of innate immunity (e.g. THP-1, MUTZ-3, MM6) will be assessed alone and - to emulate the intestinal *in vivo* environment- also in combination with human stem cell derived intestinal epithelial cells (IECs). Effects on cell differentiation and on modulation of immune related pathways in IECs and immune cells will be evaluated using whole genome gene expression and cytokine profiling. These effects will be compared with effects observed using PBMCs and relevant subsets, and the gut epithelium used in Task 3.4, ensuring connectivity between models used.
Task 3.4: Allergenicity risk assessment toolbox, Duration M4-45 (**INRAE**, WR, UMan, MUS, ROQ, BD, DNR). The phylogenetic relationships between species from which novel proteins are derived and those characterised as being allergenic (Task 3.1) will be described, to guide the assessment process (at whole food level). The capacity

of existing *in silico* tools to identify risks relating to cross-reactive allergies will be assessed and a novel 3D modelling approach to define specific HLA binding motifs will be implemented to improve predictive capacity. This will be complemented by a simplified serum screening microfluidic method using foods identified in Task 3.1. A suite of *in vitro* cell lines and *ex-vivo* samples will be assessed with regards to their utility to provide data to assess de novo sensitization risks posed by alternative proteins. Methods and readouts (e.g. changes in expression of surface markers, cytokines) will be developed using the hyper- and hypo-allergens and their digests and compared with the alternative protein ingredients (Task 3.2) using the following models:

(1) Gut epithelium (INRAE) models delivered by WP4 will be adapted to mimic either a healthy or an atopic, proinflammatory phenotypes (e.g. treatment with IL25/33, TSLP and/or 'challenged' to increase permeability, e.g. via DON, Tox-A, histamine);
(2) Antigen Presenting Cells (WR) based on cell line-derived (e.g. THP-1) dendritic cells (DCs) will be applied to identify which parts of (novel) proteins (immunogenic epitopes) will be processed by DCs into their MHC-II complex. Subsequently, such 'primed' DCs will be assessed for their capacity to activate naive PBMC-derived T-cells, to create a Th2-skewing micro-environment, via analysis of surface markers and cytokine expression;
(3) B-cells (UMan) using either established cell lines (e.g. Namalwa) or isolated from HLA-typed human volunteers (healthy, and atopic individuals with and without food allergy accessed through the ManARTS Biobank). Cell-lines and human donors will be HLA typed and these data, together with published human HLA data sets, used to model the binding of peptides (MUS) generated either by in vitro digestion (Task 3.2) or intracellular (endosomal) processing using MS (UMan, UNa). This will identify digestion-resistant peptides that can act as immunogens. The outputs will be compared with an assessment of de novo sensitization in volunteers in the WP4 DigestAA trial to determine food specific plasma IgGs, IgE, IgA and IgM (INRAE).

Task 3.5: Post-market risk communication and surveillance of alternative proteins. Duration: M12-48 (**DAAB**, UMan, FSN). A stakeholder workshop (clinicians, food industry, analytical sciences, risk assessors and risk managers) will be held (DAAB) to consider options for informing consumers of risks posed by novel foods (including the phraseology on food labels), using allergen advisory allergen labelling for novel foods as an example. Workshop outputs will be used to adapt the iFAAM survey tool by the food allergic patients (in DE and UK) to discover (a) how well the issues of cross-reactive allergens and novel protein ingredients are understood, and (b) evaluate the suggested phrases and how useful these are in supporting patient safety.
This will form the basis for an information strategy on the use of labelling of cross-reactive allergens and a short guide to allergens and alternative protein ingredients accessible by consumers, healthcare professionals and the food industry. This will be complemented through the development of a pilot *Allergic Reactions Reporting Observatory* by adapting the iFAAM on-line-reactions in the community tool to run a public platform e.g. REDCap portal from iFAAM to collect alerts of adverse reactions from patients and the clinical community (in DE and UK).

Deliverables

D3.1 Risk assessment scenarios for alternative protein ingredients with safety by design approach, UMan (M18)
D3.2 In-silico and novel in-vitro methods to identify compounds of toxicological concern, WR (M40)
D3.3 Integrated *in silico* and *in vitro* test tools for assessing allergenic risks, INRAE (M45)
D3.4 Report on cross-reactive allergen labelling with implications for stakeholders, and an e-Observatory of allergic reactions in the community, DAAB (M48).

Work package No.	4	**Lead beneficiary**					TG	
Work package title	Digestibility and health							
Participant No	7	11	12	13	26	28	30	32
Participant	UNa	TG	INRAE	APT	UL	FC	ROQ	DNR
Person months:								
Start month	1					**End**	42	

Objectives

O4.1 Establish a nutritional passport for alternative protein ingredients and derived foods, and monitor their fate during *in vitro* gastro-intestinal transit and absorption of nutrients
O4.2 Develop novel and improved *in vitro* methods for determination of protein quality and bioavailability
O4.3 Determine digestibility of proteins by an *in vivo* human intervention trial
O4.4 Assess the down-stream health effects of alternative proteins, their ingredients and formulated foods

Description of work

Task 4.1 *In vitro* assessment of digestive behaviour. Duration: M1-42 (**TG**, UNa, UL, FC, ROQ, DNR)
Alternative protein ingredients and derived products from WP2 will be assessed for their digestive behaviour *in vitro* methods. Information on the digestive behaviour and ANF will be used to adjust the functional properties and processing methods of the ingredients in WP2 (TG, UNa). The following analyses will be undertaken:

a) Nutritional AA balance and protein digestibility: screening of ingredients and formulated foods from WP2 using recognised methods, such as the ISO/INFOGEST[31] method for adults, replacing traditional animal models (DIAAS); Results will be compared with currently DIAAS tested proteins or other networks.
b) Digestive behaviour: Selected foods, especially those with available digestibility data, will be digested using more sophisticated (semi-)dynamic methods[32] for in-depth digestive analysis, e.g. slow/fast and early/late release of nutrients, food matrix interactions, process-induced modulation of (micro)-structure, gastric restructuring etc.
c) Monitoring and identification of anti-nutritional factors (ANF) in alternative protein products by *in vitro* digestion before and after BBM to evaluate peptide and AA profiles. As part of this task, current animal BBMs will be replaced with *in vitro* BBMs. This will include: a) comparison of porcine and Caco2 cell BBM enzymes for composition, abundance and activity; b) evaluation of batch-to-batch variation of activity for standardisation purposes, isolation efficiency, and activity of peptidases; c) integration and validation of the new INFOGEST model in collaboration with international networks i.e. ring trials, and uptake by industry (UL, FC, ROQ, DNR).

Task 4.2 Bioavailability of (alternative) proteins. Duration: M3-30 (**TG**, UNa, UL, FC, ROQ, DNR)

First, parameters of a healthy Caco2-HT29MTX co-culture as a gut barrier model will be defined, including a ring trial of 5 laboratories from WP3 and 4. Once a consensus protocol is agreed upon, the new model will be further elaborated to accommodate events in the gut lumen i.e. food digesta. This accommodation will incorporate digesta detoxification recommendations from INFOGEST. This work will culminate in detailing the permeability, across our gut barrier model, of amino acids, di- and tripeptides released from test (alternative) proteins. Permeability of other nutritional components in test foods which can be influenced by proteins will also be tracked.

Task 4.3 *In vivo* assessment of digestive behaviour. Duration: M1-42 (**APT**, UNa, TG, INRAE)

In vivo digestion of **two selected alternative protein-based foods** (selection based on results from WP1, 2, 3, 5 and 7), will be assessed in two groups of human healthy volunteers, thanks to intestinal sampling (n=8/ group; DigestAA study). Volunteers will be equipped with a naso-intestinal tube the day before the postprandial test.[33] When the tube is located at the ileal site, samples will be collected continuously for 8h after the ingestion of the test meal. The protein and amino acid digestibility/bioavailability from the two foods will be determined in the ileal samples, using isotope labelling. Two strategies will be used:

1. Direct quantification by labelling the food product with ^{15}N; Undigested dietary protein and amino acids will be quantified in the ileum using Isotope Ratio Mass Spectrometry (IRMS) coupled with Elemental Analyser or Gas Chromatography-combustion (GC-C). For a complete overview of protein quality, metabolic bioavailability of dietary nitrogen will be determined by quantifying deamination losses.
2. Indirect quantification by labelling endogenous protein via orally administered $^{2}H2O$ on the evening before the test.[34] The determination of ^{2}H enrichment in individual amino acids allows discriminating between endogenous (^{2}H labelled) and dietary (unlabelled) amino acids. This will enable the determination of the bioavailability of individual amino acids. This method is pioneering and will be used complementary to or as an alternative to the ^{15}N labelling as proof of concept to improve methodology for digestion studies

Task 4.4 Health and wellness biomarkers Duration: M28-42 (**TG**, UNa, UL, FC, ROQ, DNR)

'Health and wellness' features of alternative proteins will be determined. The gut barrier model will be used in the presence of GI digested proteins. Tight junction integrity and architecture, mucus production, cytokine secretion and redox balance will be monitored. Alternative proteins will also be rated post-gastric and post-intestinal digestion for their ability to modulate satiety hormones in STC-1 cells. STC-1 will also be seeded on the apical face of the collected gut barrier and basolateral samples for downstream satiety signalling and insulin secretion assays. To assess the ability of test proteins to promote growth and/or maintenance of muscle, C2C12 myoblasts will be treated with samples collected from Caco2-HT29MTX basolateral chambers.

Deliverables

D4.1 Nutritional passports of alternative protein ingredients and foods, covering digestive behaviour, co-nutrients, safety and potential health benefits, TG (M42)
D4.2 Report on *in vivo* digestibility of two alternative protein foods, INRAE (40)
D4.3 Report on down-stream health effect of alternative protein ingredients and derived foods, TG (M42)

Work package No.	5	**Lead beneficiary**		MATIS	
Work package title	Sustainability and climate				
Participant No	3	8	9	10	21	33
Participant	MATIS	LUKE	WU	DIL	GCL	ETHZ
Person months:						
Start month	1			**End**	48	

[31] Brodkorb, A. et al. 2019. doi:10.1038/s41596-018-0119-1
[32] Mulet-Cabero, A.I. et al. (2020). doi: 10.1039/C9FO01293A
[33] Calvez, J. et al. (2020) doi: 10.1093/ajcn/nqaa274
[34] Tessier, R. et al. (2020) doi: 10.1093/jn/nxz279

Objectives

O5.1 Estimate the overall sustainability of the production of selected alternative proteins through environmental, economic and social LCA, and compare with traditional protein sources
O5.2 Assess the appropriateness of selected alternative proteins to serve as climate mitigation and/or adaptation solutions
O5.3 Offer direction for improvements and sustainable growth within the selected alternative protein systems to promote improved sustainability, circularity potentials, resilience and climate neutrality
Environmental Life Cycle Analysis (E-LCA), Environmental Life Cycle Costing (E-LCC) and Social Life Cycle Assessment (S-LCA) will be integrated into a **GIANT LEAPS sustainability framework**, also including impacts on the five main drivers causing global biodiversity loss through a hybrid LCA/non-LCA methods.

Description of work

Task 5.1 System mapping and sustainability framework including PEF. Duration: M1-24 (**ETHZ**, LUKE, WU, DIL). The sustainability of the value chains of the selected alternative proteins (short list) will be mapped, including (i) identification of key actors associated with the major stages; (ii) identification of key impact categories (all partners) to be explored in a quantitative way; (iii) economic potentials (WU) based on qualitative research regarding market forecasts for product penetration; and (iv) qualitative illustration of how and what side-streams are important to alternative protein production. A framework (ETHZ) linking alternative proteins to the product environmental footprint (PEF) method will be developed. This will be accomplished via a literature review to determine the current state of best practices for assessing the sustainability of alternative proteins. Finally, based on this, an **integrated GIANT LEAPS sustainability framework** will be proposed.
Task 5.2 Inventory data collection protocol and data gathering. Duration: M1-36 (**DIL**, LUKE, GCL, ETHZ). A unified system will be developed for data integration in the Life Cycle Inventory (LCI) data repository (DIL). This repository will be used for the sustainability impact assessment Task 5.3. Data collection protocols will be designed for efficient and consistent data collection, fulfilling the goals and sustainability framework (Task 5.1) and requirements of sustainability assessment (Tasks 5.3). The data collection system will be set (DIL) in accordance with WP6 and include consideration of types, formats, sources, and fluxes. Next, data from literature, public funded projects, proprietary databases, project partners and AB members and other stakeholders will be collected using the developed protocol (LUKE, GCL, ETHZ). Aggregation and anonymization procedures will be applied to avoid potential conflicts of interests and inconsistencies. The LCI will be structured through a modular system (modules representing reference unit balance models) defined according to the functional units, system boundaries, allocation/substitution procedures. Social impacts and risks data will be collected and integrated into the joined LCI framework based on the recommendations from UNEP, Life Cycle Initiative and Social LC Alliance. All data will be integrated from available databases and from partners in other WPs.
Task 5.3 Sustainability assessment. Duration: M12-46 (**LUKE**, DIL, WU, GCL). The environmental, social and economic aspects of alternative protein production chains will be assessed. The results of the three sustainability pillars will be integrated (LUKE, GCL) to provide a full sustainability analysis for the short list of alternative proteins and selected derived foods. *E-LCA assessment* will be applied to assess the environmental sustainability (LUKE) of the whole production chain of the alternative proteins and to identify the environmental hotspots in their production. New PEF based categories will be created for the selected novel proteins. *S-LCA assessment:* The Social LCA framework[35] will be used (LUKE) to assess social risk hotspots and potential benefits in all relevant stakeholder groups. A special focus will be paid to pathways for a just and fair transition to healthier and more sustainable proteins. *LCC assessment:* A Life Cycle Costing approach (WU, DIL) will be used to assess the economic sustainability of the production chains of alternative protein sources. As externalities are assessed, this assessment focuses on the internal costs of the production chain to evaluate its economic viability.
Task 5.4 Climate adaptation and mitigation. Duration: M12-M46 (**MATIS**, LUKE).
Alternative proteins as climate mitigation and adaptation solutions: Results from the *E-LCA* will be used to estimate (LUKE) the potential of alternative proteins as mitigation solutions, as low greenhouse gas emitters compared to traditional animal proteins. Alternative proteins' potential as adaptation strategies will be explored.
Climate vulnerability assessment and adaptation strategies: Product vulnerability will be assessed (MATIS) for selected (short list) alternative protein production processes. First, exposure and sensitivity towards climate change impacts will be estimated throughout the production cycles, including markets and shipment. Second, the adaptive capacity of the production cycle will be estimated considering social, economic and environmental aspects.
Task 5.5 Biodiversity and Ecosystem services. Duration: M12-M38 (**LUKE**, MATIS)
LCA and non-LCA biodiversity assessment: The impacts related to biodiversity will be assessed considering the five main drivers causing global biodiversity loss (habitat change, climate change, invasive species, overexploitation and pollution) identified by the Millennium Ecosystem Assessment[36]. An LCA method will be

[35] Benoît, C. et al (2010). The international journal of life cycle assessment, 15(2), 156-163.
[36] Duraiappah AK et al (2005). Ecosystems and human well-being; a report of the Millennium Ecosystem Assessment.

used to cover three of the drivers, and separate (non-LCA) indicators will be used to assess the impacts caused by the other two drivers (invasive species and overexploitation), for which no operational LCA-methods yet exist.
Ecosystem services of alternative proteins: The most important ecosystem services related to the production of the alternative protein sources will be identified and impacts on them will be assessed qualitatively, in case no quantitative method is available. The impact of natural capital loss in all regions under study will be assessed as to inform local stakeholders. Results will feed into the overall sustainability assessment in T5.3.
Task 5.6 Circularity potential. Duration: M24-M45 (**DIL**, MATIS, WU, ETHZ)
Circularity potential through end-of-life waste treatment substitution: This task will estimate (DIL) the potential level of waste biomass inclusion for the cultivation/growth of alternative proteins based on the product level circularity index, further adapted for food systems. As the standardized approach for the economy level circularity is under development (ISO/WD 59020.2), the task will rely on global, regional and national nutrient flows and budgets (MATIS, WU, ETHZ) to estimate the circular economy potential of alternative proteins on higher levels. A multi-level approach will determine potential benefits and drawbacks in LCA and sustainability scenarios.
Circularity potential through environmental impact allocation to (re-usable) by-products: Potential utilization of side and by-products of alternative protein sources for applications in food and non-food systems will be identified, from techno-environmental and economic (all task partners) perspectives following the food eco-design strategies (design for recycling and upcycling, design for reuse, design for sustainability).

Deliverables

D5.1 Manual for the Life Cycle Inventory database, DIL (M36)
D5.2 Report on the sustainability and biodiversity of production of alternative proteins, LUKE (M46)
D5.3 Report on the appropriateness of alternative proteins to fight against climate change, MATIS (M46)
D5.4 Report on the circular potential of producing alternative proteins, DIL (M45)

Work package No.	6	**Lead beneficiary**			DIL	
Work package title	Data integration platform					
Participant No	10	15	16	21		
Participant	DIL	AZTI	IRIS	GCL		
Person months:						
Start month	1				**End**	48

Objectives

O6.1 Set up a data integration platform ensuring interoperability and data access to stakeholders
O6.2 Collect existing and generated data and integrate these in the open cloud platform
O6.3 Develop explainable Artificial Intelligence (AI) algorithms for predicting protein properties and their potential for sustainable integration in diets

Description of work

Task 6.1 Data integration framework and interoperability ontology. Duration: M3-24 (**DIL**, AZTI, IRIS, GCL)
Protocol parameters for the procedure of data integration (available in different databases and collected in WP1-5 and 7) will be defined (DIL, IRIS, AZTI). The formatted protocols, such as GS1 GDSN and/or FoodEx2, will be shared with the consortium and Stakeholder Board members for further data collection and integration (Task 6.2). Evidence-based scientific data (available following FAIR principles) will be integrated through unified ontologies from all WPs (IRIS, DIL) and external sources (GCL). Ontologies built on existing developments (e.g., FoodOn) will be further used to define the links between the different types of semantic knowledge which help in formulating the data search strategies, useful for data users with different needs. A framework and ontology for data interchange and integration will be created and crosslinked with other digital Farm-to-Fork projects (such as FNS-Cloud and SmartCLIDE) with further integration in EOSC and JRC databases.
Task 6.2 Development of cloud architecture platform. Duration: M1-48 (**IRIS**, DIL, AZTI)
The back end, database, end-points APIs, user hierarchy, front-end and technical specifications for database selection and consensus protocols will be designed, and the external connections for modular platform exploitation will be defined. Conceptual diagrams and methodologies for each use case will be established, which will achieve a novel interoperable data platform for protein-relevant factors (explored in WP1-5, 7), since there is a need for an improvement in data representation to facilitate the processing by machine learning algorithms.
Task 6.3. Explainable AI method for predictive protein screening. Duration: M14-48 (**AZTI**, DIL, IRIS)
A 'predictive protein screening' algorithm, targeting identification of potential interdependencies between environmental, nutritional and safety properties of alternative proteins (traditional proteins, long list and beyond) will be developed (AZTI) that combines black-box models (neural networks or support vector machine) with white box methods (explainable AI methods: supervised learning techniques), 'active learning' or ensemble methods (Random Forest, XGBoost). Statistical modelling techniques, e.g. PCA will also be explored to test the output explainability of a possible hybrid model (AI and statistical model combined). The algorithms will be trained with

the cloud platform database and, therefore, adapted to data size, type and structure of stored datasets. Successful algorithms for protein screening will be selected and integrated in the cloud platform (DIL, IRIS). The code of the AI algorithms will be made openly available on GitHub.

Deliverables

D6.1 Data integration and interoperability ontology report, DIL (M24)
D6.2 User manual for GIANT LEAPS cloud data platform, IRIS (M36)
D6.3 Predictive explainable AI algorithms for screening of protein sources integrated in platform, AZTI (M48)

Work package No.	7		**Lead beneficiary**				WR			
Work package title	Optimising the dietary shift									
Participant No	1	3	5	8	9	15	26	27	28	33
Participant	WR	MATIS	VTT	LUKE	WU	AZTI	UL	CAP	FC	ETHZ
Person months:										
Start month	1						**End**	45		

Objectives

O7.1 Assess the impacts of the foreseen dietary shifts using alternative protein derived foods on human health and the environment, and compare these with current, traditional diets based
O7.2 Arrive at accepted optimal dietary shifts, accepted by consumers and complying to health and environmental requirements

Description of work

Task 7.1 Composition of futures diets with alternative protein foods. Duration: M1-12 (**WU**, WR, VTT, AZTI, UL, CAP, FC, ETHZ)
Current diets (with large amounts of animal proteins) and future diets with alternative protein derived foods will be composed for each of four regions in Europe (North, South, West and East). Based on available data from the EFSA food consumption database (at FoodEx level), the typical current diets will be composed; this will be an 'average' diet comprising 3-4 countries per region, per consumer group (children, adults etc). Based on these current diets, future diets will be composed with alternative protein foods as part of our regular daily meals to replace animal proteins, considering (national) dietary guidelines, cultural differences and typical consumption habits in the particular regions (all). Input from WP1 will be used to assure future diets are accepted by consumers.
Task 7.2 Identification of indicators and data collection for health and environment. Duration: M1-24 (**WR**, WU, MATIS, LUKE)
Indicators for use in the models (Task 7.3) will be established considering the four impact groups of safety (WR), allergenicity (WR), nutrition (WU) and environmental impacts (LUKE, MATIS). Each impact group can have multiple indicators. Data on these indicators will be collected for both the alternative protein derived foods and the traditional foods in WP2 to 6. In this task, data collection will be completed by literature reviews, expert interviews, and on-going (inter)national projects, e.g. the H2020 mEATquality project. Data will be formatted for delivery to WP6 for storage in the GIANT LEAPS cloud infrastructure, and for use in Tasks 7.3 and 7.4.
Task 7.3 Impacts of dietary shift on health and environment. Duration: M15-42 (**WU**, WR, ETHZ)
The SHARP (Sustainable, Healthy, Accessible, Reliable and Preferred) model (https://www.tifn.nl/sharp/), which includes effects of dietary patterns on all macro-nutrients and several micro-nutrients, as well as on GHG and land use for the traditional foods, will be used as starting point. It will be extended (WU) with data on N and P, and water use, and for alternative protein derived foods. Using the extended model, the impacts of dietary patterns on nutritional intake, the environment, safety and allergenicity (WU) will be estimated. This will be done for the current and future diets (Task 7.1), for each of the four European regions, and per consumer group. Data from WP5 and 6, and Task 7.2 will be used, at FoodEx level. Using the extended SHARP model, synergies and trade-offs between health and environment will be investigated. Trade-offs will also be assessed by considering the opinions of the different stakeholder groups, using multi-criteria decision analyses (MCDA) (WR). Stakeholder preferences will be retrieved by a survey (WR) among different groups of stakeholders across Europe, during the 3rd co-creation workshop of WP1, and by an additional consumer survey in the four regions (ETHZ). MCDA allows for relatively quick scenario analyses of impacts of diets on the four indicator groups. Scenario and sensitivity analyses will be done (WR).
Task 7.4 Optimization of dietary shift. Duration: M24-45 (**WR**, VTT, WU, AZTI, UL, CAP, FC, ETHZ)
Multi-objective optimization (WR) will be used to arrive at 'optimal' dietary shifts, based on the established indicators for health and environment, their weights and constraints. Constraints are derived from legal frameworks, task partners involved, and from stakeholder visions, obtained from WP1 and 8. The optimal diets can mitigate environmental impact while at least keeping health related indicators similar to current diets.

Furthermore, they will take into account maximum amounts of alternative proteins and cultural differences between the European regions. Based on the results, together with Task 7.3 results, policy recommendations will be formulated (all), which feed back for discussion in the last workshop with stakeholders in Tasks 1.1 and 8.3.

Deliverables

D7.1 Review paper on indicators and data for dietary impacts on health and environment, WR (M24)
D7.2 Report with impacts of dietary shifts on health and environment, compared with traditional diets, WU (M42)
D7.3 Policy recommendations, with pros and cons of the dietary shift, WR (M45)

Work package number	8											**Lead beneficiary**			EFF		
Work package title	Communication, dissemination and exploitation																
Participant No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Participant	WR	IRTA	MATIS	UCC	VTT	UMan	UNa	LUKE	WU	DIL	TG	INRAE	APT	MUS	AZTI	IRIS	EFF
Person months:																	
Participant No	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
Participant	B2F	VM	SF	GCL	EM	FSN	Mosa	NFB	UL	CAP	FC	DAAB	ROQ	BD	DNR	ETHZ	AGT
Person months:																	
Start month	1											**End**		48			

Objectives

O8.1 Enable effective communication and dissemination of GIANT LEAPS and results
O8.2 Maximize the exploitation of GIANT LEAPS products and innovations
O8.3 Develop recommendations for policymakers
O8.4 Educate and train professionals and EU citizens

Description of work

Task 8.1 Effective communication and dissemination. Duration: M1-48 (**EFF**, all partners)

A **communication and dissemination strategy plan**, visual identity and website will be developed, which includes (i) creation of a dissemination strategy (with key messages and target audiences) (EFF), planning of communication and dissemination activities (EFF), (iii) creation of a **visual identity** (logo, typography, graphic elements, branding guidelines and templates) (EM), and (iv) creation of a **specific website** (EM). The website will be set-up and maintained (EM) for the duration of the GIANT LEAPS project and kept online for 3 years after the completion of the project. The website will be integrated with the e-learning platform of Task 8.4 (EM), allowing to follow the learning curve of interested participants. E-learning for children and wider public will use games.

Dissemination materials will be produced and distributed. Materials include(i) two project leaflets (EFF) (one presenting the project's main elements, and one summarising the project's main achievements), (ii) one infographic and two animation videos (EFF) (one on project introduction, and one on project results), (iii) social media presence (EM/B2F), (iv) regular (each 6 months) newsletters (EFF), (v) popular articles (EFF) and (vi) 9 policy briefs (FSN) (developed in Task 8.3). **Dissemination events** will be organised and participated in to ensure a broad uptake of GIANTLEAPS results at regional, national and international level, including (i) attending dissemination events organised by parties from outside (e.g. BBI and EIT) and inside the consortium (e.g. the yearly B2F workshops) (all partners), (ii) organization of 4 online webinars (EFF), (iii) 12 presentations at scientific conferences organised by parties outside and partners inside the consortium (e.g. the annual EFFoST conference) (all partners), and (iv) organization of specific GIANT LEAPS conference at the end of the project (B2F).

Task 8.2 Set up exploitation pathway, and plan for exploitation and dissemination of project results. Duration: M1- 48 (**EFF**, B2F, EM, FSN)

An **exploitation pathway** will be designed to capture all exploitable results (EFF, B2F), which includes (i) definition of an appropriate IPR strategy, (ii) assessment of the market context to be equipped to face market barriers and capitalise on drivers, and (iii) quantification of the addressable market to set targets for each solution. Exploitable project results will be identified with WP leaders on a bi-yearly basis. A detailed **exploitation plan** will be designed covering all relevant project results (EFF, B2F), including (i) identify the most appropriate business models for the project results and innovations, (ii) develop business plans for commercial solutions, and (iii) determine the steps required to bring the newly developed technologies to TRL9 after the end of the project. It will be combined with the communication and dissemination strategy plan of Task 8.1 to form the Plan for the **Exploitation and Dissemination of project Results (PEDR)** including communication activities.

Task 8.3 Recommendations for policymakers. Duration: M1-45 (**FSN**, EFF, B2F, EM)

Interactive face-to-face and on-line sessions with a range of policymakers at the EC level, including EFSA, at

individual Member States, and at the global (WHO/FAO) level will be held. The face-to-face sessions will be organized, in line with the three WP1 co-creation workshops, whereas the online sessions will be held with smaller groups of policymakers. Input from the policymakers on the protein transition and constraints will be communicated to the consortium for consideration in WP work, and vice versa. Ultimately, this will result in a series of 9 **Policy Briefs** on a variety of topics of relevance, containing information on the advantages and disadvantages of the alternative protein foods and dietary shifts on health and the environment, fully supported by science and presented in tangible format for policymakers to digest. Also, the knowledge produced through all WPs will be discussed with policymakers in a separately organized **workshop with EU policymakers** (M42).

Task 8.4 Education and training. Duration: M24-45 (**EM**, EFF, B2F, FSN, all)

Two GIANT LEAPS training programmes, aimed to support the protein transition, will be developed and offered (EM); one to professionals (scholars, students, researchers, interested parties from the industry, health-care professionals, and policymakers etc.) and one to EU citizens (lay(wo)men). Training materials will include a set of training modules that will also be made available online to be followed as an e-learning program. Content for the training program comes from all partners, EM will transform these inputs into a game-based e-learning program.

Deliverables

D8.1 Plan for Exploitation and Dissemination of project Results (PEDR) including communication activities, EFF (M6, updated M18, 30, 42)
D8.2 Exploitation pathway, EFF (M48)
D8.3 Report with policy briefs and policy workshop outcomes, FSN (M45)

Work package number	9		**Lead beneficiary**				WR	
Work package title	Coordination							
Participant No	1	3	5	6	10	11	17	33
Participant	WR	MATIS	VTT	UMan	DIL	TG	EFF	ETHZ
Person months:								
Start month	1					**End**	48	

Objectives

O9.1 To manage the progress of the project to the timely completion of the milestones and deliverables, at the highest possible quality level and within the planned resources (operational level)
O9.2 To manage the IPR of the project innovation and results, Data Management, and fulfilling ethical requirements, and manage project risks and unexpected situations (strategic level)
O9.3 To manage the consortium's relationship with the EC by ensuring that all the required documentation is delivered according to the given rules and procedures (organizational level)

Description of work

Task 9.1 Operational project management. Duration: M1-48 (**WR**, MATIS, VTT, UMan, DIL, TG, EFF, ETHZ)

This task includes all activities to ensure that the project Deliverables and Milestones are achieved within time, budget, quality and scope; therefore, this covers all aspects relating to daily project management for implementing the project, including project monitoring and control, managing financial and administrative issues, organising of project and committees' meetings, preparing the periodic reports to the EC etc

The **Project Coordinator** (PC), Dr AP Vos of WR, who has wide experiences in leading research in food industry and NGO, will take the leading role in these activities to ensure the contractual agreements with the EC are met. He will be assisted by an Administrative Project Manager and the Deputy Coordinator (both WR), together forming the **WR coordination team**. The **Deputy Coordinator**, Prof. HJ van der Fels-Klerx, has wide experience in leading (inter)national projects and she will assist the PC whenever needed, and replace the PC in the event of any unforeseen occasions. The WR coordination team will be assisted by the WR financial, ethics and legal department. The WR coordination team, together with the WP leaders and the IIPR manager, form the **Executive Committee** (ExC). The WR coordination team will organize the meetings, together with involved partners, including 5 face-to-face annual meetings with the entire consortium, and 4 face-to-face annual meetings with the ExC.

Task 9.2 Strategic project management. Duration: M1-48 (**WR**, MATIS, VTT, UMan, DIL, TG, EFF, ETHZ)

Internal and external risks, and unexpected situations occurring within the project will be systematically assessed and once identified, managed by the PC together with the ExC. These could be any internal or external risks jeopardizing the project's implementations, its scientific and technological outcomes, or be of economical, IPR, ethical or legal character. A well-known example is the COVID crisis which led to temporary closure of working areas and laboratories in 2019 and 2020. Risks and their mitigation strategies have already been identified prior to the projects start (Table 3.2e) and will be updated every 6 months by the ExC.

The PC will make sure GIANT LEAPS complies as much as possible to the principles of **Open Science and Data Management**. An experienced **Open Science and Data Manager (OSDM)** (WR) will be appointed for this task;

she/he will also train the involved partners to comply with the Open Science and Data Management plan. The OSDM, assisted by all partners, will elaborate a Data Management Plan (DMP), following the format provided by Horizon Europe (see section 1.2.8 "Research data management and management of other research outputs" for more details). The DMP, delivered at M6, will be updated every 6 months.

Management of Intellectual Property Rights (IPR): **a project Innovation and IPR (IIPR) manager** (WR) will be appointed from the WR Legal department (which has a special group on IPR issues). The IIPR manager will evaluate the results and innovations from GIANT LEAPS, as early as possible, for their novelty, and possible patentability and protection of the results under fair conditions between the involved partners. Also, project partners will inform the IIPR manager and PC upon identification of possible IPR. Where an IPR (e.g. patent) application is required, it will be filed and reported (D9.2)

Task 9.3 Organization of project management. Duration: M1-48 (**WR**)

The PC is the linking point between the consortium and the European Commission (EC) project officer. The PC will manage the consortium's relationship with the EC by ensuring that all the required documentation is obtained according to the given rules and procedures. The PC will prepare, with input from all partners, the project's **periodic technical and financial reports** (M18, 36) and the **final technical and financial report** (M48). As part of this reporting, **ethical and gender** aspects will be detailed (M18, 36, 48).

Deliverables

D9.1 Data Management Plan (DMP), WR (M6)
D9.2 Intellectual Property Rights report, WR (M45)
D9.3 Ethics and gender report, WR (M48)

Table 3.2c: List of Deliverables

Deliverable (number)	Deliverable name	WP No.	Short name of lead participant	Type	Dissemination level	Delivery date (months)
D1.1	Reports of 3 co-creation workshops	1	B2F	R	PU	3, 24, 48
D1.2	Promising alternative proteins (foods) to consumers	1	ETHZ	R	PU	24
D1.3	Consumer trade-offs and dietary shifts policies	1	ETHZ	R	PU	46
D2.1	Characterisation of alternative protein ingredients	2	MATIS	R	PU	12
D2.2	Functionalisation strategies for alternative ingredients and food models	2	UCC	R	SEN	24
D2.3	Alternative protein ingredient based innovative food structures	2	VTT	R	SEN	36
D3.1	Risk assessment scenarios for alternative protein ingredients with safety by design approach	3	UMan	R	SEN	18
D3.2	In-silico and novel in-vitro methods to identify compounds of toxicological concern	3	WR	DEM	PU	40
D3.3	Integrated *in silico* and *in vitro* test tools for assessing allergenic risks	3	INRAE	DEM	PU	45
D3.4	Cross-reactive allergen labelling, stakeholder implications, and allergenic reactions e-Observatory	3	DAAB	R	PU	48
D4.1	Nutritional passports of alternative protein ingredients/foods, incl. digestive behaviour, co-nutrients, safety and potential health benefits	4	TG	R	PU	42
D4.2	*In vivo* digestibility of 2 alternative protein foods	4	INRAE	R	SEN	36
D4.3	Down-stream health effects of alternative protein ingredients and foods	4	TG	R	SEN	42
D5.1	Manual for the Life Cycle Inventory database	5	DIL	OTHER	PU	36
D5.2	Sustainability and biodiversity of production of alternative proteins	5	LUKE	R	PU	46
D5.3	Appropriateness of selected alternative proteins to serve in the fight against climate change	5	MATIS	R	PU	46
D5.4	Circular potential of producing alternative proteins	5	DIL	R	PU	45
D6.1	Data integration and interoperability ontology	6	DIL	OTHER	PU	24
D6.2	GIANT LEAPS cloud data platform user manual	6	IRIS	OTHER	PU	36

D6.3	AI model algorithms integrated in gamma (final) version of open cloud platform	6	AZTI	OTHER	PU	48
D7.1	Health and environmental indicators with data for traditional and alternative protein-based diets	7	WR	R	PU	24
D7.2	Impact of the future dietary shift on health and environment, compared with traditional diets	7	WU	R	PU	42
D7.3	Policy recommendations with pros and cons of the dietary shift	7	WR	R	PU	45
D8.1	Plan Dissemination and Exploitation of results (PEDR) including communication activities	8	EFF	R	PU	6
D8.2	Exploitation pathway	8	EFF	R	PU	48
D8.3	Policy briefs and policy workshop outcomes	8	FSN	R	PU	45
D9.1	Data Management Plan (DMP)	9	WR	DMP	PU	6
D9.2	Intellectual Property Rights	9	WR	R	SEN	45
D9.3	Ethics and gender report	9	WR	ETHICS	PU	48

Table 3.2d: List of milestones

Milestone No.	Milestone name	Related WP(s)	Due date (month)	Means of verification
1	Strategy for GIANT LEAPS stakeholder network beyond Europe and project	1	36	Strategy approved by the PC and WP leaders
2	Consumer preferences for alternative proteins (foods) available to WPs 2 and 7	1	18	Consumer survey data results sent to WP 2 and WP7 leaders
3	Alternative protein ingredients for characterization, modification and testing	2	2	Ingredient supply by industrial partners (AGT, CN, GCL, BD, Solar) to WP2
4	Ingredient processing data	2	24	Data provided to WP5 and 6
5	Protein case studies selected	3	4	Case studies approved at the first ExC meeting
6	Ethical approvals submitted for eObservatory	3	18	Ethics submission reference and associated documents
7	*In vitro* digestion screening of eight protein products or ingredients	4	18	Digested samples collected and preserved for further analysis
8	*In vitro* gut barrier models treated with eight proteins products or ingredients	4	24	Caco2-HT29MTX basolateral samples collected for downstream cellular assays
9	Completion of an international ring trial to define a healthy in vitro gut barrier	4	28	Identification of three potential protein products for human study, based on stakeholder feedback; feasibility of isotope production established
10	*In vitro* methods to predict the digestive behaviour and absorption of novel proteins	4	42	International consensus
11	Framework for sustainability assessment	5	18	Framework approved by ExC and WP5 partners
13	Protocols for data collection for the Life Cycle Inventory data repository	5	26	Protocols approved by ExC and WP5 partners
15	Protocol for collection and ontology for data integration	6	12	Protocol and ontology approved by ExC and WP6 partners
16	Alpha version of GIANT LEAPS cloud data platform	6	18	Alpha version available for testing by project partners
17	Current and future diets composed for each of the four European regions	7	12	Future diets presented at 2nd project meeting and approved by partners
18	Indicators for estimating dietary shift impacts on health and environment	7	12	Indicators presented at 2nd project meeting and approved by partners
19	Project logo and website	8	3	Logo approved by partners, and website available to the public
20	Dissemination materials ready	8	12, 24, 36, 48	Dissemination materials sent to partners for distribution
21	E-training module	8	32	E-training module approved by ExC
22	Project mid-term review	9	22	Review held and report available

Table 3.2e: Critical risks for implementation

Description of risk (level of (i) likelihood, and (ii) severity: Low/Medium/High)	WP(s) involved	Proposed risk-mitigation measures
Lack of engagement/interest of stakeholders (med/med).	1	GIANT LEAPS partners will use their extensive network to reach selected stakeholders. The approach, tools and messages will be aligned to the needs of the stakeholders.
Specific groups of stakeholders refuse to participate in joint workshops (med/med).	1, 8	Separate workshops will be organized per stakeholder group, separate interaction in WP8 with policymakers
Changes in raw material occur during supply or storage of the raw materials (low/med)	2	Raw materials will be delivered and stored under appropriate conditions. Fresh raw material batches will be used in case of inappropriate storage
Challenges in improving the critical functional properties (sensory, technological) (med/med)	2	Protein modification will be used in WP2 to tailor functional properties. The project has several end-use applications with different target functionalities.
Lack of consistency in application of in vitro digestion methods gives inconsistent results between partners (low/high)	3	Prepare harmonised detailed protocols and SOPs and undertake pilot analysis to ensure comparability.
Delays in obtaining ethical approval for the on-line (questionnaire based) allergic reactions observatory (low/high)	3	Study is based on a similar study in the past (iFAAM) for which ethical approval was obtained. Draft study protocol and address sponsorship and data management issues as soon as the project commences.
Insufficient volumes of serum and cells obtained from sufficient number of human study subjects (low/high)	3	WP3 uses historic serum samples available at partners and samples from two different tissue banks. Recruitment of study subjects in centres with large patient cohorts. Begin in year 2 to ensure sufficient time for recruitment.
Too many protein ingredients and foods for digestion screening (high/med)	4	Regular revision of list of protein products by the ExC ranking and prioritising of proteins
Insufficient number of volunteers for human intervention trial (med/high)	4	Early start of planning (ethics etc.) from day 1; decision of two protein sources by Month 12; work with world leaders in the field of digestion human intervention trials
Lack of reliable data for assessment (med/high)	5	Multiple sources are used, and aggregation procedures are established. Data collection procedures for missing aspects will be provided and integrated in the analysis.
Limited ability to collect primary data for social sustainability due to movement and meeting restrictions (low/low)	5	Face-to-face focus meetings remain the best approach for data collection. Great advances have been done in remote connections, hence very little risk of not gathering data.
Inconsistent data quality from various sources	6	Data aggregation procedures applied to data from multiple sources. Data quality will be defined and communicated.
Lack of data with primary quality (directly measured data with confirmed origin)	6	Data imputation methods will be applied to extrapolate values.
EU-COVERAGE of current protein intake data does not reflect actual EU- diversity of dietary habits, hindering modelling accuracy (low/medium)	7	Diet survey data on baseline protein intake from all four major EU regions will be used. For these data, additional EU-countries will be explored to maximise applicability of results, via exploiting existing networks within the project.
Poor availability/accessibility of data on indicators (low/high)	7	Close collaboration with WP6 and consultation with data experts ensures use of most relevant and up-to-date indicator databases
Lack of technology acceptance by consumers will hinder market introduction (medium/medium)	8	Consumers (associations) will be targeted in dissemination and communication plan and awareness will be created early in the process
Competing patents have already been granted (low/ medium)	8	Regularly monitor patent databases and adjust work plan if necessary.
Conflicts between partners	9	The consortium agreement will describe duties and rights of each organisation and procedures to solve conflicts
Overseen cost items/expenses	9	Close follow-up of budget in shorter time-periods

Table 3.2f: Summary of staff effort

Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	Total PM
1 / WR										
2 / IRTA										
3 / MATIS										
4 / UCC										
5 / VTT										
6 / UMan										
7 / UNa										
8 / LUKE										
9 / WU										
10 /DIL										
11 / TG										
12 / INRAE										
13 / APT										
14 / MUS										
15 / AZTI										
16 / IRIS										
17 / EFF										
18 / B2F										
19 / VM										
20 / SF										
21 / GCL										
22 / EM										
23 / FSN										
24 / Mosa										
25 / NFB										
26 / UL										
27 / CAP										
28 / FC										
29 / DAAB										
30 / ROQ										
31 / BD										
32 / DNR										
33 / ETHZ										
34 / AGT										
Total PM:										

3.2 Capacity of participants and consortium as a whole

Consortium description, complementarity and industry involvement

GIANT LEAPS has mobilized the key scientists across Europe, together representing the **entire alternative protein value chain, from farm to fork**, and has set up **co-creations among all relevant bodies in Europe**, involving consumers, NGOs, producers and processors, and governmental agencies, in all stages of the research project to assure working on the right topics, and acceptance and implementation of project results.

The GIANT LEAPS consortium consists of:

- 34 leading partners from 20 different countries, of which 31 countries across Europe and 4 associated and affiliated countries (UK, Canada, Switzerland, Serbia).
- From total of 34 partners, 50% (17) are universities and research institutes (WR, IRTA, MATIS, UCC, VTT, UMan, UNa, LUKE, WU, DIL, TG, INRAE, APT, MUS, AZTI, GCL, ETHZ), and the other 50 % (17) are industries, of which 11 SMEs and start-ups (IRIS, B2F, VM, SF, EM, FSN, Mosa, NBT, CAP, DAAB, BD) and five leading larges food industries (AGT, UL, FC, ROQ, DNR) and one other partner (EFF).

In preparing the proposal, the consortium partners together with the involved members of the **Stakeholder Board** (primary producers, food industry, policymakers, consumers, NGOs etc) have **identified the main areas to be addressed** for providing the science-base for producing and using more alternative protein derived foods in our daily diets, for all stakeholders in the value chain, from farm to fork. **Exploitation of GIANT LEAPS results and innovations and their future implementation in practice** in industry is ensured by the large involvement (50%) of SME and large industries in our consortium, and the actors represented in the Stakeholder Board.

The composition of the consortium reflects the involvement of **key actors with complementary types of expertise and competences, both from science and practice** (from SME to large food industries) on all topics necessary to reach the overall project's aim, including multi-actor engagement and stakeholder involvement, consumer research, protein processing, allergenicity and health, environmental, climatic and biodiversity, and diet modelling. The co-creation approach and multi-actor partnership enables the inclusion of the latest technological developments, and existing knowledge and experience throughout the entire project, to provide science-based information to policymakers, consumers, and other stakeholders. Also, it enables to generate innovative solutions that are more likely to be applied in practice thanks to the cross-fertilisation of ideas between science and industry, to move up the innovations with one or two TRL levels, from 3-4 to 5-6.

Multiple national and international projects focus(ed) on alternative protein sources, as indicated in Table 1.4. This proposal brings together all the separate pieces of existing information and data, and addresses all missing pieces of the puzzle. **Consortium members are involved in many of the previous and ongoing EU projects** that have a link to the protein transition. For instance, the consortium includes the **four coordinators of the ongoing Horizon 2020** projects funded under topic "LC-*SFS-17-2019: Alternative proteins for food and feed*", namely SmartProtein (ECC), NextGenProteins (MATIS), ProFuture (IRTA) and SUSINCHAIN (WR). Expertise in social sciences is broadly represented (consumer research: ETHZ; social/economic LCA: LUKE, WU, DIL; stakeholder engagement:B2F).

Contribution of each partner to the project with a valid role and with the necessary resources

Partners are listed in table 3.2 for each of the core research areas around which the WPs are centred, showing that the expertise and involvement of each partner is relevant, duplications are avoided, and no key expertise is missing.

Table 3.2. Core role of each partner to the key domains of GIANT LEAPS

WP	Key expertise / domain	Key partners	Contributing partners
1	Multi-actor approach, stakeholder and scientific network, and consumer studies	ETHZ, B2F, EFF	WR, IRTA, MATIS, UCC, VTT,FSN, Mosa, CAP
2	Alternative protein processing and characterization	VTT, UCC, IRTA	MATIS, VM, SF, GCL, NFB, UL, FC, BD, AGT
3	Food safety and allergenicity	UMan, WR, INRAE, DAAB	VTT, Una, TG, MUS, FSN, Mosa, UL, DAAB, ROQ, BD, DNR
4	Digestibility (in vitro and in vivo) of alternative proteins	TG, UNa, INRAE	TG, APT, Mosa, UL, FC, ROQ, DNR
5	Sustainability, climate and biodiversity	MATIS, LUKE, WU, DIL	GCL, ETHZ
6	Open data platform and data collection	DIL, IRIS, AZTI	GCL
7	Dietary intake modelling (WR, WU)	WR, WU	MATIS, VTT, LUKE, AZTI, UL, CAP, FC, ETHZ
8	Training, dissemination, communication	EFF, B2F, EM	All partners
9	Coordination	WR	All WP leaders

Each partner is experienced in working on national and international research projects (all partners already had a PIC number) and in a multi-disciplinary setting. They are used to **planning research, with the necessary resources needed to fulfil the planned activities**. The distribution of budget among the partners follows their planned activities in the project (see Table 3.1f: Summary of staff effort) to make sure that all planned project work will be done in time, with high quality and within the budget to complete the tasks in a strict timeframe. For each partner, personnel costs are based on average staff costs, calculated from actual costs of the different groups of personnel (e.g. engineers, senior scientist) involved in the activities of the respective partner. A **high number (at least 17) of PhDs** will be involved in this project, supervised by the experienced senior scientists and professors of our consortium. In this way, GIANT LEAPS will contribute to training young scientists, ensuring our future academic staff in Europe.

Physical travels and subsistence have been minimized to **five (annual) consortium meetings** for all partners, of which Stakeholder Board members will attend three meetings (kick-off, mid-term and final), and the **three co-creation workshops** will be held in conjunction with these three meetings. Four physical half- yearly meetings will be held with the Executive Committee. In this way, the number of physical travels is limited to minimise the project's footprint, yet keeping the important interaction moments in the project. Working in a multidisciplinary setting, at the top level of research across different European cultures, needs a minimum of interactions for exchange of knowledge, progress and planning. **Many online meetings** are planned, in smaller or larger settings. WP meetings will be held online every 3 months, involving partners in respective WPs. Interactions with the Stakeholder Board will be done, whenever needed, in bilateral contacts or in smaller virtual groups. All partners have budgeted for **diverse goods and services**; these mainly involve a budget for publishing in Open Journals and subscription to conferences and events, for early and open dissemination of project results, other Open Science and Data Management activities.

All partners have full access to the **necessary personnel and infrastructure** to carry out the activities, directly and for five partners also via subcontracting. Details on infrastructure can be found in the Part A. Highlights are:

- WR has **high-end analytical equipment and state-of-the-art facilities for** safety and allergenicity research.
- IRTA has state-of-the-art pilot scale equipment for **food processing. food preparations and sensory** rooms.
- UCC has pilot scale malting, brewing and bakery including all required apparatus like **milling and mixture equipment's and rheology equipment and kitchen with sensory labs**.
- VTT has **dry and wet fractionation technology and extrusion** technologies
- UMan has facilities for all allergy related research including a **biobank of patient materials, immune and flow cytometry equipment and hospital settings** for patient tests. UMan will collaborate intensely with UNa because of **high-end NMR and complementary MS facilities required for the allergy research**.
- WU and WR have large access to **datasets, models and software tools related to health, nutrition and sustainability** assessment, supporting evaluations of dietary shift scenarios.
- DIL have all **LCA and DSS tools and software** in house for the environmental footprint analysis and more.
- TG is fully equipped for the **digestion research with spectroscopy and microscopy**, and state-of-the-art pilot plan food processing facilities, from large scale spray dryers to small scale microfluidizers.
- INRAE has excellent facilities for digestion research based on health volunteers and patients including a **fully equipped human research unit**.
- MUS is specialised in **high performance computing service and software for data storage and modelling.**
- AZTI is specialised in data **platforms, deep learning servers, computation clusters and software** related to nutrition. IRIS has complementary **software for data management, machine learning, real time monitoring** of processes and modules for building a SaaS platform. These companies will interact and collaborate with CAP, which has complementary **software for diet assessments and food composition databases** and tools.
- EFF and B2F are international network organisations that connect with large and **diverse sets of stakeholders** (e.g. from research, industry and policymakers) with the **capacity to organise** meetings, workshops and events.
- EM is highly complementary because of the **web-based solutions, illustrators and text writers** that are highly experienced to support efficient communication cross Europe stakeholders.
- Many of the involved food-oriented SME have their specific equipment and pilot scale facilities for **biomass production, processing, product development and product testing** leading to many innovative ingredients, processes, solutions and products. These SME can and will be supported by the larger food industrial partners like UL, FC, ROQ, DNR that have high-end in-house technology and expertise to **optimise and upscale processes and product development** and implement the knowledge in product innovation.
- Our consortium includes several large research organizations, such as WR, WU, VTT and INRAE, which as a fall-back scenario, have the personnel and infrastructure for the project's research activities.

All GIANT LEAPS partners fully acknowledge Ethic and GDPR regulations. The GIANT LEAPS consortium has an **equal gender distribution**, with more **than > 58% females** involved in the research activities, 33% females as main contact points and 30% in leading roles like WP leaders. The Deputy Coordinator (prof. HJ van der Fels-Klerx, and the WP2 (prof. Nesli Sözer) and WP3 (prof. Clare Mills) leaders are excellent leading female scientists. Most of the partners are in the process of implementing a formal Gender Equality Plan or already have one in place.

3.2.1 Other countries and international organisations

ETHZ from Switzerland is a leading partner and scientific authority on consumer studies. Switzerland has been classified as 'a non-associated third country' for all Horizon Europe calls in 2021, however, their **funding is guaranteed** by the Swiss State Secretariat for Education, Research and Innovation (SERI). As such, ETHZ will participate in GIANT LEAPS with budget for their activities provided by SERI (if GIANT LEAPS is granted), implying budget to fund ETHZ is not part of the requested EC contribution. AGT, the large food industrial partner from Canada, will contribute to the project with in-kind contribution, unique expertise and ingredients. UMan from the United Kingdom and CAP from Serbia join as partners from third countries associated to Horizon Europe.